A single kinetic defect in a run-and-tumble ring: exact spectrum, exceptional points, and relaxation crossover

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Abstract. We solve the spectral problem for one continuous-time run-and-tumble particle on an n-site ring when the tumbling rate at one site is ω_0 and the bulk rate is ω. The defect changes the $2n \times 2n$ generator by rank at most one. The characteristic polynomial is the homogeneous polynomial plus $(\omega_0 - \omega)/n$ times its ω-derivative. This identity gives Chebyshev and transfer-matrix quantization formulas, a parity factorization, and finite polynomial tests for every Jordan block. In the general affected sector, the exceptional locus is specified implicitly by a finite resultant equation, with root orders determining its Jordan blocks. We separately resolve the singular flat-band point $\omega = 1$, where the eigenvalue -2 can carry Jordan blocks of sizes two or three. Off the bulk spectral set, the infinite-line Green function gives an exact criterion for exponentially localized defect modes, including a separately evaluated compact mode at the singular transfer point. For fixed positive bulk rate, reflection protects one homogeneous long-wave mode; consequently a slow defect, including $\omega_0 \asymp n^{-1}$, does not produce a new spectral-gap scaling. The nontrivial n^{-1} crossover occurs when the bulk rate itself is of order n^{-1}. Finally, at fixed ω the standard telegraph equation misses the lattice Poisson-diffusion term—the componentwise $\frac{1}{2}\partial_x^2$ contribution generated by unit-rate Poisson hopping—while a diffusion-corrected telegraph equation agrees through order n^{-2}.

Keywords. Run-and-tumble particle; kinetic defect; non-Hermitian spectrum; exceptional point; localized mode; spectral gap; telegraph equation.

1. Introduction

Persistent random walks provide a microscopic description of motion with a finite directional memory. Continuum random-walk formulations of biased chemotactic motion arise, for example, in [Sch93], while lattice run-and-tumble models retain the individual hopping and reversal events and connect directly to nonequilibrium particle systems [TTCB11]. For a homogeneous ring, the one-particle spectrum, its exceptional points, and the associated relaxation transition are known explicitly [MBE19]. Spatial inhomogeneity has been studied through position-dependent speed [AG19], quenched disorder in speed and tumbling [BWK+19, Secs. 2.1.2–2.1.3 and 2.2.2–2.2.3], and saturating space-dependent rates on the line [JC24, Sec. II, Eqs. (3)–(4)]. Direction- and position-dependent reversal rates on the line are treated in [SSK20, Secs. 2–4]. Those works emphasize probability laws, steady states, or relaxation in extended media. Impurity-activated reversals on a periodic exclusion lattice have also been used to obtain exact interacting-particle phase behavior and, in a special parameter regime, a mapping to many run-and-tumble particles [CH23, Secs. V–VII]. A moving site at which tumbling is certain provides a time-periodic drive in a persistent exclusion process [DC25, Secs. II–IV]. Those models concern interacting-particle steady states and driven clustering rather than the full spectrum of a fixed one-particle tumbling-rate defect. Persistent lattice walks with site disorder have also been studied through random asymmetric transmittances [SM08]. For nonpersistent confined lattice walks, exact propagators admit Chebyshev representations; the same framework treats partially absorbing defects and connects to pseudo-Green functions [Giu20, Secs. VI–VII and Apps. C and E]. Difference-equation methods for isolated lattice impurity levels go back to the two linear-chain calculations of Koster and Slater [KS54, abstract]. Green-function reduction of a point perturbation has also been used for non-Hermitian lattices, notably to turn a periodic chain into an open one [Roc21, Secs. II–III and Eqs. (11)–(15)]. Exact finite-size impurity spectra and size-dependent localization have been obtained in nonreciprocal chains [LZLC21], while recent Green-function work treats bound states and exceptional points for complex impurities in finite and infinite non-Hermitian lattices [KKME26]. Those tight-binding problems concern on-site potentials, boundary sensitivity, and skin or bound-state localization. The present problem instead asks for the full finite-ring characteristic polynomial and Jordan structure of a single lattice tumbling defect, together with its large-ring spectral ordering.

This paper asks how that spectral picture changes when the tumbling rate is altered at one site. The inhomogeneity is local but the generator is non-normal, so eigenvalue crossings, Jordan structure, and long-wave relaxation must be tracked together. The central observation is that the defect is rank one in the polarization channel. It leads to a finite characteristic identity, a parity decomposition, an infinite-line Green-function criterion including a compact defect mode, and a direct comparison between lattice and telegraph relaxation. The fixed-rate and bulk-critical limits separate the effects of a local defect from those of a vanishing bulk tumbling rate.

The comparison above also delimits the contributions established here. The cited homogeneous spectral solution supplies the Fourier eigenvalues, homogeneous exceptional points, and first-mode relaxation transition; the cited inhomogeneous run-and-tumble works supply probability laws, steady states, and first-passage or extended-medium relaxation results; and the confined-walk reference supplies nonpersistent propagators and defective trapping formulas. Beyond those results, the present defect problem yields the rank-one derivative identity and parity factorization, the exact finite algebraic exceptional-point classification including the tuned J_3 flat-band block, the localized-mode criterion and compact tuning, and the uniform fixed-rate and bulk-critical gap selection. The continuum comparison additionally constructs a point-defect telegraph analogue at unit site spacing and isolates the lattice Poisson-diffusion correction. The point interaction is not asserted as an operator limit under a vanishing lattice spacing.

The main logical dependencies are summarized in Table 1. In particular, the gap conclusions use the root-exhaustion results, not only local expansions at the first pole.

The proof is organized by its consumers. Theorems 3.3 and 3.4 give the exact finite

Supplier	Direct consumers
Fourier blocks and the rank-one defect	characteristic, Chebyshev, transfer, and parity formulas
Parity polynomials and pole coincidences	regular and flat-band Jordan classification
Finite Green identity	infinite-line localization and fixed-rate root separation
Critical local, endpoint, and off-pole Green bounds	critical right-edge exhaustion and the double-scaling gap
Exact homogeneous dispersion	telegraph and diffusion-corrected telegraph comparison

Table 1: Dependency spine of the theorem package.

characteristic equations; Theorems 4.8 and 4.9 give the Jordan classification; Theorem 5.3 and Proposition 5.4 treat localized modes; Theorems 6.7 and 6.17 determine the two large-ring gap regimes; and Proposition 7.4 gives the independently defined continuum-analogue point-defect equation.

2. Model and notation

Definition 2.1 (Defective run-and-tumble generator)**.** Let $n \geq 2$, let $\omega > 0$, and let $\omega_0 \geq 0$. Sites are $j \in \mathbb{Z}/n\mathbb{Z}$, orientations are $\sigma \in \{+, -\}$, and

$$\omega_j = \begin{cases} \omega_0, & j = 0, \\ \omega, & j \neq 0. \end{cases}$$

The forward generator $L = L_n(\omega, \omega_0)$ is

$$(Lu)^+_j = u^+_{j-1} - (1 + \omega_j)u^+_j + \omega_j u^-_j, \tag{2.1}$$
$$(Lu)^-_j = u^-_{j+1} - (1 + \omega_j)u^-_j + \omega_j u^+_j. \tag{2.2}$$

Thus hopping in the current orientation has rate one and tumbling has rate ω_j. Put

$$\delta = \omega_0 - \omega, \qquad q_k = \frac{2\pi k}{n}, \qquad c_k = \cos q_k, \qquad s_k = \sin q_k.$$

Equations (2.1)–(2.2) use the same one-particle convention as equations (6)–(10) of [MBE19]. That paper records the homogeneous eigenvalues in its equation (10), the homogeneous exceptional points in equation (13), and the first spatial-mode relaxation rate in equations (20)–(21).

Lemma 2.2 *(Rank-one defect)***.** *Let $L^{(0)} = L_n(\omega, \omega)$. If $d = e_{0,+} - e_{0,-}$, then*

$$L = L^{(0)} - \delta\, dd^{\mathsf{T}}.$$

In particular, the perturbation has rank one when $\delta \neq 0$.

Proof. At site zero, changing ω to ω_0 adds

$$\delta\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix} = -\delta\begin{pmatrix}1\\ -1\end{pmatrix}\begin{pmatrix}1 & -1\end{pmatrix}.$$

Every other matrix entry is unchanged. ■

3. Three exact characteristic equations

Definition 3.1 (Homogeneous factors)**.** For $\lambda \in \mathbb{C}$, set

$$\begin{aligned} a &= \lambda + 1 + \omega, && (3.1)\\ x &= \frac{a^2 + 1 - \omega^2}{2a}, && (3.2)\\ D_k(\lambda) &= (\lambda + 1 - c_k)(\lambda + 1 + 2\omega - c_k) + s_k^2 = a^2 + 1 - \omega^2 - 2ac_k. && (3.3)\end{aligned}$$

Although x is undefined at $a = 0$, each expression below containing $a^n T_n(x)$ is understood by its unique polynomial continuation.

Lemma 3.2 *(Homogeneous polynomial)***.** *The homogeneous characteristic polynomial is*

$$\begin{aligned} P_n^{(0)}(\lambda;\omega) &= \det(\lambda I - L^{(0)})\\ &= \prod_{k=0}^{n-1} D_k(\lambda) && (3.4)\\ &= 2a^n\big(T_n(x) - 1\big), && (3.5)\end{aligned}$$

where T_n is the Chebyshev polynomial of the first kind. The two roots of D_k are

$$\lambda_k^{\pm} = c_k - 1 - \omega \pm \sqrt{\omega^2 - s_k^2}.$$

Proof. Fourier transformation gives the block

$$W_k = \begin{pmatrix} e^{\mathrm{i}q_k} - 1 - \omega & \omega\\ \omega & e^{-\mathrm{i}q_k} - 1 - \omega\end{pmatrix}.$$

Its characteristic polynomial is D_k, which proves (3.4) and the displayed roots. Since $D_k = 2a(x - c_k)$ and

$$\prod_{k=0}^{n-1}(x - c_k) = 2^{1-n}\big(T_n(x) - 1\big),$$

equation (3.5) follows. ■

Theorem 3.3 *(Exact finite characteristic polynomial)***.** *For every $n \geq 2$, $\omega > 0$, and $\omega_0 \geq 0$,*

$$\begin{aligned} P_n(\lambda;\omega,\omega_0) &:= \det(\lambda I - L_n(\omega,\omega_0)) && (3.6)\\ &= P_n^{(0)}(\lambda;\omega) + \frac{\delta}{n}\,\partial_\omega P_n^{(0)}(\lambda;\omega) && (3.7)\\ &= P_n^{(0)}(\lambda;\omega)\left[1 + \frac{2\delta}{n}\sum_{k=0}^{n-1}\frac{\lambda + 1 - c_k}{D_k(\lambda)}\right]. && (3.8)\end{aligned}$$

Equation (3.8) is interpreted by polynomial continuation at a homogeneous eigenvalue. Equivalently,

$$P_n \;=\; 2a^n(T_n(x)-1)+2\delta a^{n-1}(T_n(x)-1)+\delta a^{n-2}\lambda(\lambda+2)U_{n-1}(x), \tag{3.9}$$

where U_{n-1} is the Chebyshev polynomial of the second kind and the right-hand side again denotes its polynomial continuation.

Proof. Let $B \;=\; \lambda I - L^{(0)}$. Lemma 2.2 and the matrix determinant lemma give

$$\det(\lambda I - L) \;=\; \det B\bigl(1+\delta d^{\mathsf{T}}B^{-1}d\bigr).$$

In Fourier space,

$$d^{\mathsf{T}}B^{-1}d \;=\; \frac{1}{n}\sum_{k=0}^{n-1}\frac{2(\lambda+1-c_k)}{D_k(\lambda)},$$

which proves (3.8). Direct differentiation of (3.4) gives

$$\partial_\omega P_n^{(0)} \;=\; P_n^{(0)}\sum_k \frac{2(\lambda+1-c_k)}{D_k},$$

and hence (3.7). Finally,

$$\partial_\omega x \;=\; \frac{\lambda(\lambda+2)}{2a^2}, \qquad T_n'(x) \;=\; nU_{n-1}(x),$$

and differentiating (3.5) proves (3.9). ∎

Theorem 3.4 *(Transfer-matrix quantization)***.** *For a local rate $r \ge 0$, define*

$$T_r(\lambda) \;=\; \frac{1}{\lambda+1+r}\begin{pmatrix} 1 & r \\ -r & (\lambda+1+r)^2-r^2 \end{pmatrix}.$$

Then $\det T_r \;=\; 1$, and, away from the displayed denominators,

$$P_n(\lambda;\omega,\omega_0) \;=\; a^{n-1}(\lambda+1+\omega_0)\left\{\operatorname{tr}\bigl[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\bigr]-2\right\}.$$

After multiplication by the denominators, this identity holds for every λ by polynomial continuation. When $a \neq 0$ and $\lambda+1+\omega_0 \neq 0$, an eigenvalue is equivalently characterized by

$$\operatorname{tr}\bigl[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\bigr] \;=\; 2.$$

At a zero of either denominator, the cleared polynomial identity (3.4), rather than the undefined trace in (3.4), is the eigenvalue criterion. Moreover, if $t \;=\; \operatorname{tr} T_\omega \;=\; 2x$, then

$$T_\omega^{n-1} \;=\; U_{n-2}(x)T_\omega - U_{n-3}(x)I,$$

with $U_{-1} \;=\; 0$.

Proof. The eigenvalue equations at site j are

$$\begin{aligned}(\lambda + 1 + \omega_j)u_j^+ - \omega_j u_j^- &= u_{j-1}^+,\\ (\lambda + 1 + \omega_j)u_j^- - \omega_j u_j^+ &= u_{j+1}^-.\end{aligned}$$

Consequently

$$\begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix} = T_{\omega_j}(\lambda)\begin{pmatrix} u_{j-1}^+ \\ u_j^- \end{pmatrix}.$$

Periodic boundary conditions require the product monodromy to have eigenvalue one. Its determinant is one, so this is equivalent to trace two. To identify the polynomial, retain temporarily arbitrary rates $r_0, \ldots, r_{n-1}$, denote the resulting generator by $L_n(r_0, \ldots, r_{n-1})$, and put $b_j = \lambda + 1 + r_j$. Let E_j^+ and E_j^- denote the two equation residuals, with their right-hand sides moved to the left. For $b_j \neq 0$, replace each row pair by

$$\begin{pmatrix} F_j^+ \\ F_j^- \end{pmatrix} = \begin{pmatrix} b_j^{-1} & 0 \\ -r_j b_j^{-1} & -1 \end{pmatrix}\begin{pmatrix} E_j^+ \\ E_j^- \end{pmatrix}.$$

This block row operation has determinant $-b_j^{-1}$. If $y_j = (u_j^+, u_{j+1}^-)^\mathsf{T}$, the transformed equations are exactly

$$y_j - T_{r_j}(\lambda)y_{j-1} = 0, \qquad j \in \mathbb{Z}/n\mathbb{Z}. \tag{3.10}$$

Let A be the original coefficient matrix, F the matrix after these row operations, and C the matrix of (3.10) with columns ordered as $y_0, y_1, \ldots, y_{n-1}$. The column reordering cyclically sends $(u_0^-, u_1^-, \ldots, u_{n-1}^-)$ to $(u_1^-, \ldots, u_{n-1}^-, u_0^-)$ and fixes the plus variables, so its sign is $(-1)^{n-1}$. Therefore

$$\det F = \frac{(-1)^n}{\prod_j b_j}\det A, \qquad \det C = (-1)^{n-1}\det F.$$

Successively eliminating $y_0, \ldots, y_{n-2}$ in (3.10), using its identity diagonal blocks, leaves $(I - M)y_{n-1} = 0$, where $M = T_{r_{n-1}} \cdots T_{r_0}$; hence $\det C = \det(I - M)$. Combining the three determinant identities gives

$$\det A = -\prod_j b_j \det(I - M) = -\prod_j b_j \det(M - I).$$

Since $\det M = 1$, one has $-\det(M - I) = \operatorname{tr} M - 2$. Consequently

$$\det(\lambda I - L_n(r_0, \ldots, r_{n-1})) = \left(\prod_{j=0}^{n-1} b_j\right)\left[\operatorname{tr}(T_{r_{n-1}} \cdots T_{r_0}) - 2\right]. \tag{3.11}$$

This is an identity of rational functions where all $b_j \neq 0$. The displayed product clears every denominator on the right, giving a polynomial, so the identity extends to every λ. Taking $r_0 = \omega_0$ and all other rates equal to ω, and using cyclicity of the trace, gives (3.4), including algebraic multiplicities. Equation (3.4) is the Cayley–Hamilton identity for a unimodular 2×2 matrix. ∎

4. Parity factorization and every finite exceptional point

Definition 4.1 (Kinetic reflection)**.** Define the involution

$$(Ru)_j^+ = u_{-j}^-, \qquad (Ru)_j^- = u_{-j}^+.$$

Write $\mathcal{H}_\pm = \ker(R \mp I)$.

Lemma 4.2 *(Protected sector)***.** *The generator commutes with R, the defect vector satisfies $Rd = -d$, and*

$$L|_{\mathcal{H}_+} = L^{(0)}|_{\mathcal{H}_+}.$$

Thus the defect changes only the reflection-odd sector.

Proof. Reflection reverses position and orientation simultaneously, leaving the directed hopping rules and the rate field ω_j invariant. At the fixed site zero it exchanges the two entries of d, hence $Rd = -d$. Lemma 2.2 then vanishes on $\mathcal{H}_+$. ∎

Definition 4.3 (Parity polynomials)**.** Put $m = \lfloor (n-1)/2 \rfloor$ and let ε_n be one for even n and zero for odd n. Define

$$H_{n,+}(\lambda) = \lambda(\lambda+2)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.1}$$

$$H_{n,-}^{(0)}(\lambda) = (\lambda+2\omega)(\lambda+2+2\omega)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.2}$$

$$G_n(\lambda) = \frac{1}{n}\left[\frac{1}{\lambda+2\omega} + \frac{\varepsilon_n}{\lambda+2+2\omega} + 2\sum_{k=1}^{m} \frac{\lambda+1-c_k}{D_k(\lambda)}\right], \tag{4.3}$$

$$Q_{n,-}(\lambda) = H_{n,-}^{(0)}(\lambda)\bigl(1+2\delta G_n(\lambda)\bigr). \tag{4.4}$$

Every quotient in (4.4) cancels, so $Q_{n,-}$ is a monic polynomial of degree n. An entirely polynomial version is

$$Q_{n,-} = H_{n,-}^{(0)} + \frac{2\delta}{n}\left[\frac{H_{n,-}^{(0)}}{\lambda+2\omega} + \varepsilon_n \frac{H_{n,-}^{(0)}}{\lambda+2+2\omega} \right.$$
$$\left. +2\sum_{k=1}^{m}(\lambda+1-c_k)\frac{H_{n,-}^{(0)}}{D_k}\right]. \tag{4.5}$$

Theorem 4.4 *(Exact parity factorization)***.** *The characteristic polynomial factors as*

$$P_n(\lambda;\omega,\omega_0) = H_{n,+}(\lambda)Q_{n,-}(\lambda).$$

The first factor is the characteristic polynomial on $\mathcal{H}_+$; the second is the characteristic polynomial on $\mathcal{H}_-$.

Proof. The modes q_k and $-q_k$ form a four-dimensional Fourier subspace, which splits into two isomorphic two-dimensional R-eigenspaces. Each has characteristic polynomial D_k. At $q = 0$, the vectors $(1,1)$ and $(1,-1)$ have parities $+$ and $-$ and eigenvalues 0 and -2ω. When n is even, the $q = \pi$ density and polarization vectors have parities $+$ and $-$ and eigenvalues -2 and $-2-2\omega$. This gives (4.1)–(4.2). Lemma 4.2 and the rank-one determinant lemma restricted to $\mathcal{H}_-$ give (4.4). Multiplication proves (4.4). ■

Definition 4.5 (Exceptional point)**.** For an eigenvalue μ, let $m_{\mathrm{alg}}(\mu)$ be its algebraic multiplicity and $m_{\mathrm{geo}}(\mu) = \dim\ker(\mu I - L)$ its geometric multiplicity. We call (ω, ω_0, μ) an exceptional point when $m_{\mathrm{alg}}(\mu) > m_{\mathrm{geo}}(\mu)$. Write $J_r(\mu)$ for a Jordan block of size r at μ.

Lemma 4.6 *(No inter-wave coincidences away from the flat band)***.** *If $c_k \neq c_\ell$ and $D_k(\mu) = D_\ell(\mu) = 0$, then $\omega = 1$ and $\mu = -2$.*

Proof. Subtracting the two equations gives $2a(c_k - c_\ell) = 0$, hence $a = 0$. Substitution into (3.3) gives $1 - \omega^2 = 0$. Since $\omega > 0$, this means $\omega = 1$ and then $\mu = -1 - \omega = -2$. ■

Lemma 4.7 *(Scalar resolvent criterion for cyclicity)***.** *Let $A \in \mathbb{C}^{N\times N}$ and $v \in \mathbb{C}^N$. Suppose the reduced denominator of the scalar rational function*

$$v^{\mathsf{T}}(\lambda I - A)^{-1}v$$

is the characteristic polynomial $\chi_A(\lambda)$. Then v is a cyclic vector for A. Moreover, for every $\eta \in \mathbb{C}$, the same vector is cyclic for $A - \eta vv^{\mathsf{T}}$.

Proof. Let $m_{A,v}$ be the monic polynomial of least degree satisfying $m_{A,v}(A)v = 0$. For $\lambda \notin \operatorname{spec} A$,

$$m_{A,v}(\lambda)(\lambda I - A)^{-1}v = \frac{m_{A,v}(\lambda)I - m_{A,v}(A)}{\lambda I - A}\, v. \tag{4.6}$$

The matrix quotient on the right is a matrix polynomial in λ: for each monomial it is the finite divided-difference identity

$$\frac{\lambda^j I - A^j}{\lambda I - A} = \sum_{r=0}^{j-1} \lambda^{j-1-r} A^r.$$

Thus the reduced denominator of every scalar component of $(\lambda I - A)^{-1}v$, and in particular of the displayed scalar resolvent, divides $m_{A,v}$. By hypothesis, χ_A therefore divides $m_{A,v}$. Conversely, Cayley–Hamilton gives $\chi_A(A)v = 0$. Write $\chi_A = q\, m_{A,v} + r$ with $\deg r < \deg m_{A,v}$. Then $r(A)v = 0$, so the defining minimality of $m_{A,v}$ forces $r = 0$. Hence $m_{A,v}$ divides χ_A, and therefore $m_{A,v} = \chi_A$ and has degree N, which is equivalent to $v, Av, \ldots, A^{N-1}v$ spanning $\mathbb{C}^N$.

Put $B = A - \eta vv^{\mathsf{T}}$. Induction on j gives

$$B^j v = A^j v + \sum_{r=0}^{j-1} c_{j,r} A^r v$$

for suitable scalars $c_{j,r}$. The change from the A-Krylov list to the B-Krylov list is therefore triangular with diagonal entries one. The two lists have the same span, so v is cyclic for B. ■

Theorem 4.8 *(Jordan classification for $\omega \neq 1$). Assume $\omega \neq 1$.*

1. *In the protected sector, the endpoint factors in (4.1) give one-dimensional blocks. For each $1 \leq k \leq m$, a root of D_k gives one J_1, except that*

$$\omega = |s_k|, \qquad \mu = c_k - 1 - \omega. \tag{4.7}$$

   gives one $J_2(\mu)$.

2. *In the affected sector, if*

$$Q_{n,-}(\mu) = Q'_{n,-}(\mu) = \cdots = Q^{(r-1)}_{n,-}(\mu) = 0, \qquad Q^{(r)}_{n,-}(\mu) \neq 0, \tag{4.8}$$

   then the complete affected-sector contribution at μ is one block $J_r(\mu)$.

3. *The full Jordan form at μ is the direct sum of the blocks in items 1 and 2. In particular, a crossing between the two parity sectors is semisimple unless one of its sector blocks has size greater than one.*

Consequently, (4.7) together with (4.8) is an exact finite criterion for every exceptional point when $\omega \neq 1$. The order r in (4.8) classifies exceptional points of arbitrary order.

Proof. The discriminant of D_k is $4(\omega^2 - s_k^2)$. At a zero discriminant its Fourier block is not scalar and hence is a single J_2; otherwise each root is simple. Lemma 4.6 excludes coincidences between distinct c_k values.

It remains to show that the affected restriction is cyclic. In each two-dimensional homogeneous parity block, the spectral weight of the defect vector has numerator $2(\lambda + 1 - c_k)$. This numerator and D_k have no common root for an interior wave number: a common root would give $\lambda + 1 = c_k$, after which $D_k = s_k^2 \neq 0$. The endpoint weights in (4.3) are also nonzero. At a double root of an interior D_k, the numerator remains nonzero, so the scalar resolvent retains a pole of order two. Regard the endpoint factors as the odd-parity roots of D_0 and, when n is even, $D_{n/2}$. Lemma 4.6 then makes their root sets disjoint from those of every interior factor and from one another when $\omega \neq 1$. Consequently no pole, including a repeated pole, cancels when the block fractions are put over their common denominator. The scalar resolvent $d^{\mathsf{T}}(\lambda I - L^{(0)}|_{\mathcal{H}_-})^{-1} d = 2G_n(\lambda)$ therefore has reduced denominator $H^{(0)}_{n,-}$, the characteristic polynomial of the homogeneous odd restriction. Lemma 4.7 makes d cyclic first for that restriction and then for its update by $-\delta d d^{\mathsf{T}}$. Hence the minimal and characteristic polynomials of the affected restriction both equal $Q_{n,-}$. A cyclic matrix has one Jordan block for each eigenvalue, of size equal to the root order. The direct-sum assertion follows from the invariant parity splitting. ■

Theorem 4.9 *(The complete flat-band classification at $\omega = 1$). Let $\omega = 1$ and put $\mu = -2$.*

1. *If* $\omega_0 = 1$ *and* $4 \nmid n$*, the contribution at* μ *is* $nJ_1(\mu)$*.*

2. *If* $\omega_0 = 1$ *and* $4 \mid n$*, the contribution is*

$$2J_2(\mu) \oplus (n-2)J_1(\mu).$$

3. *If* $\omega_0 \neq 1$ *and* $n \equiv 1$ *or* $2 \pmod 4$*, the contribution is* $(n-1)J_1(\mu)$*.*

4. *If* $\omega_0 \neq 1$ *and* $n \equiv 0 \pmod 4$*, the contribution is*

$$J_2(\mu) \oplus (n-2)J_1(\mu).$$

5. *If* $n \equiv 3 \pmod 4$*,* $\omega_0 \neq 1$*, and* $\omega_0 \neq (n-1)/(n+1)$*, the contribution is again* $J_2(\mu) \oplus (n-2)J_1(\mu)$*.*

6. *If* $n \equiv 3 \pmod 4$ *and* $\omega_0 = (n-1)/(n+1)$*, the contribution is*

$$J_3(\mu) \oplus (n-2)J_1(\mu).$$

For eigenvalues $\lambda \neq -2$*, every protected-sector root is a one-dimensional block. Remove the exact power of* $(\lambda+2)$ *specified above from* $Q_{n,-}$*. Every root of the remaining affected polynomial contributes one Jordan block, so the derivative test (4.8) applied to that reduced polynomial classifies every remaining affected block at* $\omega = 1$*. The full Jordan form is the direct sum of the protected and affected blocks.*

Proof. Write $t = \lambda + 2$. At $\omega = 1$,

$$D_k = t(t - 2c_k), \qquad P_n = t^{n-1}F_n(t),$$

where, for $\delta = \omega_0 - 1$,

$$F_n(t) = 2t\big(T_n(t/2) - 1\big) + \delta\left[2\big(T_n(t/2)-1\big) + (t-2)U_{n-1}(t/2)\right]. \tag{4.9}$$

Put $C_n = \cos(n\pi/2)$ and $S_n = \sin(n\pi/2)$. The values $T_n(0) = C_n$, $U_{n-1}(0) = S_n$, together with

$$T_n'(0) = nS_n, \qquad U_{n-1}'(0) = -nC_n,$$

give directly from (4.9)

$$F_n(0) = 2\delta(C_n - 1 - S_n), \tag{4.10}$$
$$F_n'(0) = 2(C_n - 1) + \delta\big((n+1)S_n + nC_n\big). \tag{4.11}$$

If $\delta = 0$ and $4 \nmid n$, the derivative in (4.11) is nonzero. If $\delta = 0$ and $4 \mid n$, the Chebyshev differential equation at zero gives $T_n''(0) = -n^2$, and hence

$$F_n(t) = -\frac{n^2}{4}t^3 + O(t^5).$$

For $\delta \neq 0$, equations (4.10)–(4.11) give

$n \pmod 4$	first nonzero datum	$\mathrm{ord}_{t=0} F_n$
$1, 2$	$F_n(0) = -4\delta$	0
0	$F_n'(0) = n\delta$	1
3	$F_n'(0) = -2 - (n+1)\delta$	1

in the last row unless $\delta = -2/(n+1)$. For $n \equiv 3 \pmod 4$ one has $U_{n-1}''(0) = n^2 - 1$, and direct differentiation of (4.9) at that tuned value gives

$$F_n''(0) = -2n + \frac{U_{n-1}''(0)}{n+1} = -(n+1) \neq 0. \tag{4.12}$$

Thus $\mathrm{ord}_0 F_n$ is respectively 1 or 3 when $\delta = 0$ according as $4 \nmid n$ or $4 \mid n$, and is respectively 0, 1, 1, or 2 in the four defective cases displayed above, with the final value belonging to the tuned $n \equiv 3 \pmod 4$ case. Since $P_n = t^{n-1} F_n$, this proves the stated algebraic multiplicities, including the tuned multiplicity $n + 1$.

The bulk equations at $t = 0$ read

$$u_j^- = -u_{j-1}^+ \qquad (j \neq 0).$$

They determine every minus component except u_0^- from the plus components. Put $\delta = \omega_0 - 1$. The two defect equations are

$$\delta u_0^+ - \omega_0 u_0^- = u_{n-1}^+, \qquad \delta u_0^- - \omega_0 u_0^+ = u_1^- = -u_0^+.$$

The second equation is therefore

$$\delta(u_0^- - u_0^+) = 0.$$

If $\omega_0 = 1$, the first equation supplies $u_0^- = -u_{n-1}^+$ and leaves the n plus components free. If $\omega_0 \neq 1$, the second equation gives $u_0^- = u_0^+$, and the first then gives

$$u_{n-1}^+ = (\delta - \omega_0) u_0^+ = -u_0^+.$$

Thus the geometric multiplicity is n in the homogeneous case and $n - 1$ otherwise. Fourier decomposition shows that the two excess algebraic dimensions when $4 \mid n$ and $\delta = 0$ are the two $q = \pm\pi/2$ blocks, each a J_2. For $\delta \neq 0$, the parity factorization places the $4 \mid n$ excess in one protected J_2.

It remains to distinguish one J_3 from two J_2 blocks in the tuned $n \equiv 3 \pmod 4$ case. Put $N_0 = L_n(1,1) + 2I$ and $N = L_n(1, \omega_0) + 2I = N_0 - \delta d d^\mathsf{T}$. Because $4 \nmid n$, zero is a semisimple eigenvalue of N_0 with eigenspace $E = \ker N_0$ of dimension n. Let Π be its spectral projection. Write

$$\det(tI - N_0) = t^n h(t), \qquad h(0) \neq 0.$$

Semisimplicity gives $(tI - N_0)^{-1} = t^{-1}\Pi + O(1)$. The rank-one determinant formula then shows that the coefficient of t^{n-1} in $\det(tI - N)$ is $\delta h(0) d^\mathsf{T} \Pi d$. The multiplicity computation above shows that this coefficient vanishes for $n \equiv 3 \pmod 4$, and therefore

$$d^\mathsf{T} \Pi d = 0. \tag{4.13}$$

The bulk relations derived above show directly that the eigenspace of N is $K = E \cap \ker d^{\mathsf{T}}$, which has dimension $n-1$. If $v \in K \cap \operatorname{ran} N$, write $v = Nu$ and apply Π:

$$v = \Pi N u = -\delta\, \Pi d\, (d^{\mathsf{T}} u). \tag{4.14}$$

Thus $K \cap \operatorname{ran} N$ has dimension at most one. For a finite matrix, every Jordan block of size at least two contributes its leading eigenvector to $\ker N \cap \operatorname{ran} N$, these vectors are independent, and they span that intersection. The algebraic multiplicity exceeds the geometric multiplicity here, so the intersection has dimension one. Hence exactly one Jordan block at zero for N is nontrivial. It is a J_2 when the algebraic multiplicity is n, and the additional algebraic order at $\delta = -2/(n+1)$ makes that same block a J_3. This proves all six cases and excludes the alternative of two J_2 blocks in the tuned case.

For completeness, put $A_-^{(0)} = L^{(0)}|_{\mathcal{H}_-}$ and consider an affected eigenvalue $\lambda \neq -2$. If $\delta \neq 0$, it cannot equal a nonflat eigenvalue of the homogeneous affected restriction $A_-^{(0)}$: the corresponding residue in (4.3) is nonzero, so $Q_{n,-}$ does not vanish at that pole. Hence $\lambda I - A_-^{(0)}$ is invertible. The rank-one eigenvalue equation makes its eigenspace the one-dimensional span of $(\lambda I - A_-^{(0)})^{-1} d$. Its root order therefore gives the size of its unique Jordan block. If $\delta = 0$, all nonflat roots are simple by Lemma 4.6. This proves the final classification without asserting cyclicity at the removed flat primary component. ■

Corollary 4.10 *(Finite-ring algebraic exceptional locus)***.** *For fixed n and $\omega \neq 1$, the non-flat affected exceptional parameters are exactly those satisfying*

$$\operatorname*{Res}_{\lambda}\big(Q_{n,-}, \partial_\lambda Q_{n,-}\big) = 0,$$

with $Q_{n,-}$ given by (4.5). At $\omega = 1$, put

$$r_- = \begin{cases} (n+1)/2, & \omega_0 = 1,\ n \text{ odd}, \\ n/2, & \omega_0 = 1,\ n \equiv 2 \pmod 4, \\ n/2+1, & \omega_0 = 1,\ n \equiv 0 \pmod 4, \\ (n-1)/2, & \omega_0 \neq 1,\ n \equiv 1 \pmod 4, \\ n/2-1, & \omega_0 \neq 1,\ n \text{ even}, \\ (n+1)/2, & \omega_0 \neq 1,\ n \equiv 3 \pmod 4,\ \omega_0 \neq (n-1)/(n+1), \\ (n+3)/2, & n \equiv 3 \pmod 4,\ \omega_0 = (n-1)/(n+1), \end{cases} \tag{4.15}$$

and define

$$\widehat{Q}_{n,-}(\lambda) = \frac{Q_{n,-}(\lambda)}{(\lambda+2)^{r_-}}.$$

If $n = 3$ and $\omega_0 = 1/2$, then $r_- = 3$ and $\widehat{Q}_{3,-} = 1$, so there are no non-flat affected eigenvalues. On every other parameter stratum in (4.15), the non-flat affected exceptional parameters are exactly those satisfying

$$\operatorname*{Res}_{\lambda}\big(\widehat{Q}_{n,-}, \partial_\lambda \widehat{Q}_{n,-}\big) = 0.$$

The protected exceptional parameters (4.7) and the flat-band cases of Theorem 4.9 are supplied separately; they are not asserted to arise from these affected-sector resultants. Root *orders and Theorems 4.8–4.9 distinguish exceptional points from semisimple parity crossings without a numerical tolerance. For the general affected sector, this is an implicit algebraic classification: the resultant is an exact finite polynomial equation for the exceptional locus, and the root-order tests determine the Jordan block, but no parametric solution of that resultant equation is asserted.*

Proof. For a nonconstant polynomial F, the resultant $\operatorname{Res}_\lambda(F, F')$ vanishes exactly when F has a repeated root. For $\omega \neq 1$, the polynomial $Q_{n,-}$ is monic of degree n, so Theorem 4.8 proves the first assertion. At $\omega = 1$, it remains to verify the exponent in (4.15). Definition 4.3 gives

$$\operatorname{ord}_{\lambda=-2} H_{n,+} = \begin{cases} (n-1)/2, & n \text{ odd}, \\ n/2, & n \equiv 2 \pmod 4, \\ n/2+1, & n \equiv 0 \pmod 4. \end{cases}$$

Subtracting these protected orders from the total flat algebraic multiplicities in Theorem 4.9 gives every line of (4.15). Thus $\widehat{Q}_{n,-}$ removes the exact affected flat primary factor on each stratum. At the tuned $n = 3$ point, $Q_{3,-}$ is monic of degree three and its exact flat order is three, so $\widehat{Q}_{3,-} = 1$ and the non-flat affected spectrum is empty. On every other stratum, $\widehat{Q}_{n,-}$ is nonconstant, and the resultant criterion applies. Theorems 4.8–4.9 then identify the Jordan block attached to every repeated non-flat root and separately account for the protected and flat contributions. ∎

5. Defect-localized modes

Definition 5.1 (Bulk spectral set)**.** For the infinite translation-invariant chain, define

$$\Sigma_\omega = \left\{ c - 1 - \omega \pm \sqrt{\omega^2 - (1-c^2)} \;:\; c \in [-1,1] \right\}.$$

For $\lambda \notin \Sigma_\omega$ with $a \neq 0$, retain a and x from (3.1)–(3.2), and choose

$$s(\lambda) = \sqrt{x(\lambda)^2 - 1}$$

as the standard branch on $x \in \mathbb{C} \setminus [-1,1]$ satisfying $s/x \to 1$ as $|x| \to \infty$. Equivalently, it is the branch satisfying $s(\lambda) \sim x(\lambda)$ as $|\lambda| \to \infty$. Put $z = x - s$, so $|z| < 1$.

Let $L_\infty^{(0)}$ act on $\ell^2(\mathbb{Z}; \mathbb{C}^2)$ by (2.1)–(2.2) with every local rate equal to ω, and let $L_\infty = L_\infty^{(0)} - \delta d d^{\mathsf{T}}$, where now d is the polarization vector at the origin. These operators are bounded because they are finite linear combinations of bilateral shifts and bounded multiplication operators. Infinite-line eigenvalues below always mean ℓ^2 eigenvalues of L_∞.

Lemma 5.2 *(Bulk spectrum and cyclic Green identities)***.** *The homogeneous infinite-chain spectrum is exactly*

$$\operatorname{spec} L_\infty^{(0)} = \Sigma_\omega.$$

For $x \in \mathbb{C} \setminus [-1,1]$, with $s = \sqrt{x^2-1}$ and $z = x - s$ on the branch of Definition 5.1,

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{dq}{x-\cos q} \;=\; \frac{1}{s}, \tag{5.1}$$

$$\frac{1}{n}\sum_{k=0}^{n-1} \frac{1}{x-\cos q_k} \;=\; \frac{1}{s}\frac{1+z^n}{1-z^n}. \tag{5.2}$$

The right side of (5.2) is invariant under the simultaneous branch change $(s,z) \mapsto (-s, z^{-1})$ and therefore extends meromorphically to every $x \in \mathbb{C}$ away from the finite set $\{\cos q_k : 0 \le k < n\}$, where it equals the finite sum on the left.

Proof. The unitary Fourier transform identifies $L_\infty^{(0)}$ with multiplication on $L^2([0,2\pi];\mathbb{C}^2)$ by the continuous matrix symbol

$$W(q) \;=\; \begin{pmatrix} e^{\mathrm{i}q}-1-\omega & \omega \\ \omega & e^{-\mathrm{i}q}-1-\omega \end{pmatrix}.$$

Its two pointwise eigenvalues are

$$\cos q - 1 - \omega \pm \sqrt{\omega^2 - \sin^2 q},$$

whose union is Σ_ω. If $\lambda \notin \Sigma_\omega$, then $\lambda I - W(q)$ is invertible for every q and its inverse is a continuous matrix function on the compact circle, hence is uniformly bounded and defines $(\lambda - L_\infty^{(0)})^{-1}$. Conversely, if λ is a pointwise eigenvalue at q_0, choose a corresponding nonzero vector and normalized scalar functions supported in shrinking neighborhoods of q_0. The resulting Fourier-space vectors form an approximate eigenvector sequence by continuity of W, so $\lambda \in \operatorname{spec} L_\infty^{(0)}$. This proves the spectral identity.

For (5.1), put $\zeta = e^{\mathrm{i}q}$. Since $z + z^{-1} = 2x$ and $|z| < 1$, the two roots of the resulting denominator are z and z^{-1}, and only z lies inside the unit circle. The residue theorem gives

$$\frac{1}{2\pi}\oint_{|\zeta|=1} \frac{2\mathrm{i}\, d\zeta}{(\zeta - z)(\zeta - z^{-1})} \;=\; \frac{2}{z^{-1}-z} \;=\; \frac{1}{s}.$$

Also, the absolutely convergent Poisson-kernel expansion is

$$\frac{1}{x-\cos q} \;=\; \frac{1}{s}\left(1 + 2\sum_{r=1}^{\infty} z^r \cos(rq)\right).$$

Averaging over $q_k = 2\pi k/n$ removes every term for which $n \nmid r$. Summing the remaining geometric series proves (5.2). Finally, the displayed branch change fixes its right side. It consequently defines a single-valued meromorphic function, which agrees with the finite rational sum on $\mathbb{C} \setminus [-1,1]$ and hence everywhere away from their poles. ∎

Theorem 5.3 *(Infinite-line localization criterion)*. *Let $\delta \neq 0$ and $\lambda \notin \Sigma_\omega$. First suppose $a \neq 0$. Then λ is an exponentially localized eigenvalue of the infinite chain with the same single tumbling defect if and only if*

$$1 + 2\delta g_\infty(\lambda) \;=\; 0, \tag{5.3}$$

where

$$g_\infty(\lambda) \;=\; \frac{1}{2a}\left(1+\frac{\lambda+1-x}{s}\right) \;=\; \frac{z(\lambda+1-z)}{a(1-z^2)}. \tag{5.4}$$

The localization length is

$$\xi(\lambda) \;=\; \frac{-1}{\log|z|}.$$

Equivalently, with $a \,=\, \lambda+1+\omega$, *the pair* (a,z) *satisfies*

$$a^2+1-\omega^2-a(z+z^{-1}) \;=\; 0, \tag{5.5}$$

$$a(1-z^2)+2\delta z(a-\omega-z) \;=\; 0, \qquad 0 \,<\, |z| \,<\, 1. \tag{5.6}$$

After elimination of a, *every admissible* z *is a root of*

$$\begin{aligned} 0 \;=\; &2\delta z^3+(2\delta\omega+1-\omega^2)z^2 \\ &+(4\delta^2\omega+4\delta\omega^2-2\delta)z+(2\delta\omega+\omega^2-1), \end{aligned} \tag{5.7}$$

and, when the denominator is nonzero,

$$a \;=\; \frac{2\delta z(\omega+z)}{1-z^2+2\delta z}, \qquad \lambda \;=\; a-1-\omega. \tag{5.8}$$

The zero-denominator branch in (5.8) *occurs precisely when*

$$0 \,<\, \omega \,<\, 1, \qquad \omega_0 \;=\; \frac{\omega^2+1}{2\omega}, \qquad z \;=\; -\omega. \tag{5.9}$$

At these parameters there are two admissible values

$$a_\pm \;=\; \frac{-\omega^2-1\pm\sqrt{5\omega^4-2\omega^2+1}}{2\omega}, \qquad \lambda_\pm \;=\; a_\pm-1-\omega, \tag{5.10}$$

and both give noncompact localized modes with decay root $z \,=\, -\omega$. *A root of* (5.7) *is accepted only when* (5.3), *or equivalently both* (5.5) *and* (5.6), *is satisfied; this removes roots introduced by elimination or by clearing denominators.*

The value $z \,=\, 0$, *excluded above, gives a further compact case at the point where* $a \,=\, 0$. *If* $\omega \,\neq\, 1$, *then*

$$\lambda \;=\; -1-\omega \quad \textit{is localized if and only if} \quad \omega_0 \;=\; \frac{\omega^2+1}{2\omega}. \tag{5.11}$$

Its decay root is $z \,=\, 0$, *its localization length is* $\xi \,=\, 0$ *by the finite-support convention, and its eigenvector is supported at the defect and its two neighboring sites. Thus the admissible range in the preceding formulas is* $0 \,<\, |z| \,<\, 1$; $z \,=\, 0$ *is accepted precisely in* (5.11).

Proof. The polarization entry of the bulk resolvent at the defect site is

$$\begin{aligned} g_\infty(\lambda) \;&=\; \frac{1}{2\pi}\int_0^{2\pi}\frac{\lambda+1-\cos q}{a^2+1-\omega^2-2a\cos q}\,dq \\ &=\; \frac{1}{2a}\left[1+(\lambda+1-x)\frac{1}{2\pi}\int_0^{2\pi}\frac{dq}{x-\cos q}\right] \\ &=\; \frac{1}{2a}\left(1+\frac{\lambda+1-x}{s}\right). \end{aligned}$$

Here the last equality is (5.1). If $(\lambda - L_\infty)u = 0$, then

$$u = -\delta(\lambda - L_\infty^{(0)})^{-1} d\,(d^\mathsf{T} u).$$

The scalar $d^\mathsf{T} u$ cannot vanish unless $u = 0$. Applying d^T therefore gives (5.3). Conversely, when that scalar equation holds, the displayed resolvent vector is a nonzero ℓ^2 eigenvector. Since $x = (z + z^{-1})/2$ and $s = (z^{-1} - z)/2$, elementary simplification gives the second expression in (5.4). The Fourier coefficient of $(x - \cos q)^{-1}$ is $z^{|j|}/s$; consequently, on the half-line $j \geq 0$, the second component of $(\lambda - L_\infty^{(0)})^{-1} d$ is

$$-\frac{\lambda + 1 - z}{2as}\, z^j.$$

The secular equation and $\delta \neq 0$ give $g_\infty(\lambda) = -1/(2\delta) \neq 0$. The second identity in (5.4), together with $a \neq 0$, $z \neq 0$, and $|z| < 1$, therefore gives $\lambda + 1 - z \neq 0$. This tail does not cancel, so its exact exponential rate is $|z|$ and the localization length is $-1/\log|z|$. Clearing the denominator in (5.3) gives (5.6); the bulk dispersion relation gives (5.5). After multiplication of the dispersion relation by z, the exact elimination factorization is

$$\operatorname{Res}_a\bigl(za^2 + z(1-\omega^2) - a(z^2+1),\, a(1 - z^2 + 2\delta z) - 2\delta z(\omega + z)\bigr) = z(z^2-1)\mathcal{C}_{\omega,\delta}(z),$$

where $\mathcal{C}_{\omega,\delta}$ is the cubic polynomial displayed in (5.7). The admissible noncompact range $0 < |z| < 1$ excludes the three prefactors: $z = \pm 1$ are bulk-edge points, while $z = 0$ is evaluated separately below. Thus eliminating a gives (5.7) on the stated range. Solving (5.6) for a gives (5.8).

Put $C = 1 - z^2 + 2\delta z$. If $C = 0$, the matching equation reduces to $2\delta z(\omega + z) = 0$. Since $\delta \neq 0$ and $z \neq 0$, this forces $z = -\omega$, and then $C = 0$ is equivalent to $\delta = (1 - \omega^2)/(2\omega)$. The condition $|z| < 1$ is exactly $0 < \omega < 1$. Substitution in the dispersion equation gives

$$a^2 + (\omega + \omega^{-1})a + 1 - \omega^2 = 0,$$

whose two roots are the values in (5.10). Conversely, each root satisfies both (5.5) and (5.6). At this tuning the cubic itself factors as

$$\frac{1-\omega^2}{\omega}\, z(z+\omega)^2,$$

so its double nonzero root accounts for the two values of a rather than for a repeated eigenvalue.

It remains to evaluate the point omitted by division by a. At $\lambda = -1 - \omega$ the Fourier denominator is the nonzero constant $1 - \omega^2$, and hence

$$g_\infty(-1-\omega) = \frac{1}{2\pi}\int_0^{2\pi} \frac{-\omega - \cos q}{1 - \omega^2}\, dq = \frac{\omega}{\omega^2 - 1}.$$

The secular equation is therefore equivalent to $\delta = (1-\omega^2)/(2\omega)$, which is exactly (5.11). In Fourier variables the associated resolvent vector is proportional to

$$\begin{pmatrix} -\omega - e^{-\mathrm{i}q} \\ \omega + e^{\mathrm{i}q} \end{pmatrix}.$$

Its inverse Fourier transform is supported at the defect and its two neighbors. This proves the compact case and also shows directly that it is the limiting root $z = 0$. ∎

Proposition 5.4 *(Finite-ring convergence of a localized mode)***.** *Suppose (5.3) has a simple root λ_∞ with decay root z_∞, $0 < |z_\infty| < 1$. Then, for all sufficiently large n, the finite polynomial $Q_{n,-}$ has a unique root λ_n near λ_∞, and*

$$\lambda_n - \lambda_\infty = O(|z_\infty|^n).$$

If the corresponding right eigenvector is normalized in ℓ^2, define its site mass as $|u_j^+|^2 + |u_j^-|^2$. This mass is bounded by $C|z_\infty|^{2\operatorname{dist}(j,0)}$ for a constant C independent of j and of all sufficiently large n. In the compact case (5.11), the infinite and finite eigenvalues agree exactly and the eigenvector is supported where $\operatorname{dist}(j,0) \le 1$.

Proof. Equation (5.2) applied to the bulk Green kernel gives the exact identity

$$G_n(\lambda) = \frac{1}{2a}\left[1 + \frac{\lambda+1-x}{s}\frac{1+z^n}{1-z^n}\right]. \tag{5.12}$$

Under the simultaneous branch change $(s, z) \mapsto (-s, z^{-1})$, the last two factors in the second term both change sign. The right-hand side is therefore branch invariant and continues meromorphically as the finite rational function G_n to every point away from a finite-ring homogeneous pole. This continuation statement concerns G_n only; the infinite-volume resolvent entry g_∞ remains defined on $\mathbb{C} \setminus \Sigma_\omega$. On the closed disk chosen below, which is disjoint from Σ_ω, use the common branch of s and z. Subtracting (5.4) from (5.12) then gives

$$G_n - g_\infty = \frac{\lambda+1-x}{as}\frac{z^n}{1-z^n}. \tag{5.13}$$

Choose a closed disk B about λ_∞ disjoint from Σ_ω and containing no other zero of $1 + 2\delta g_\infty$. Shrinking it if necessary, choose

$$\rho = \max_{\lambda\in B}|z(\lambda)| < r < 1.$$

By Lemma 3.2, every zero of $H_{n,-}^{(0)}$ is a finite-ring homogeneous Fourier eigenvalue and therefore belongs to Σ_ω. Hence $H_{n,-}^{(0)}$ is nonzero on B, and the zeros of $1 + 2\delta G_n$ in B are exactly the zeros of $Q_{n,-}$ there. Equation (5.13) and its λ-derivative are bounded on B by $C_B r^n$: for the derivative, the only additional factor is $n\rho^{n-1} = O(r^n)$. Rouché's theorem on ∂B, followed by the simple-zero estimate at λ_∞, gives one zero λ_n and first yields $\lambda_n - \lambda_\infty = O(r^n)$. The root equation and (5.13) then give

$$|\lambda_n - \lambda_\infty| \le C|z(\lambda_n)|^n.$$

Since $z_\infty \ne 0$ and $z(\lambda_n)/z_\infty = 1 + O(r^n)$, one has $|z(\lambda_n)/z_\infty|^n = \exp(O(nr^n)) = O(1)$. Hence $\lambda_n - \lambda_\infty = O(|z_\infty|^n)$.

Put $\rho = |z_\infty|$, $z_n = z(\lambda_n)$, and $s_n = s(\lambda_n)$. Averaging the Poisson-kernel expansion in Lemma 5.2 against $e^{\mathrm{i}jq_k}$ gives, for every integer representative $0 \le j < n$,

$$\frac{1}{n}\sum_{k=0}^{n-1} \frac{e^{\mathrm{i}jq_k}}{x(\lambda_n) - \cos q_k} = \frac{1}{s_n}\sum_{\ell\in\mathbb{Z}} z_n^{|j+\ell n|}. \tag{5.14}$$

Indeed, the Fourier coefficient on the infinite circle is $z_n^{|j|}/s_n$, and root-of-unity averaging retains exactly the indices congruent to $-j$ modulo n. Moreover, in Fourier variables,

$$(\lambda_n - W(q))^{-1}d = \frac{1}{2a_n(x(\lambda_n) - \cos q)}\begin{pmatrix}\lambda_n + 1 - e^{-\mathrm{i}q}\\ -\lambda_n - 1 + e^{\mathrm{i}q}\end{pmatrix}, \qquad a_n = \lambda_n + 1 + \omega.$$

Thus each component of the finite-ring resolvent vector is a bounded linear combination of the coefficients in (5.14) at j and $j \pm 1$. If $d = \mathrm{dist}(j, 0) \le n/2$, then the geometric series gives the exact majorant

$$\sum_{\ell\in\mathbb{Z}} |z_n|^{|j+\ell n|} = \frac{|z_n|^d + |z_n|^{n-d}}{1 - |z_n|^n}.$$

Since s_n and a_n stay away from zero and $|z_n| \ge \rho/2$ for large n, the shifts by one only alter the constant. Consequently the periodized resolvent components are bounded by

$$C\frac{|z_n|^d + |z_n|^{n-d}}{1 - |z_n|^n}, \qquad d = \mathrm{dist}(j, 0) \le n/2, \tag{5.15}$$

where C is independent of j and of all sufficiently large n. Analyticity of $z(\lambda)$ near the simple off-spectrum eigenvalue and the eigenvalue estimate already proved give

$$z_n = z_\infty + O(\rho^n), \qquad \frac{|z_n|}{\rho} \le 1 + C\rho^{n-1}.$$

It follows uniformly for $d \le n/2$ that

$$\left(\frac{|z_n|}{\rho}\right)^d \le \exp(Cd\rho^{n-1}) \le \exp(Cn\rho^{n-1}/2) = O(1).$$

For large n, $|z_n| < 1$ and $n - d \ge d$, so the wrapped term in (5.15) is no larger than its first term; also $1 - |z_n|^n$ is bounded below. Thus every component is bounded by $C\rho^d$. The local coefficients converge to a nonzero infinite-volume eigenvector, so its unnormalized ℓ^2 norm is bounded above and below by positive constants. Normalization therefore changes the component bound only by a fixed factor, and squaring the two components proves the site-mass bound $C\rho^{2d}$.

In the compact case, the Laurent-polynomial Fourier vector displayed in the proof of Theorem 5.3 is an exact finite-ring eigenvector for every n (with the two neighbors identified when $n = 2$), proving the last assertion. ∎

6. Large-ring spectral gap

Lemma 6.1 *(Irreducibility)***.** *For every $n \ge 2$, $\omega > 0$, and $\omega_0 \ge 0$, the directed transition graph of $L_n(\omega, \omega_0)$ is strongly connected. Consequently zero is an algebraically simple eigenvalue of $L_n(\omega, \omega_0)$.*

Proof. Choose a bulk site $r \neq 0$. Starting from any state, repeated hops in its current orientation reach r. At r the orientation can be retained or reversed, the latter transition having rate $\omega > 0$. Repeated hops in the selected orientation then reach any prescribed target site with the prescribed orientation. Thus every state communicates with every other state, including when $\omega_0 = 0$.

For every $t > 0$, each such finite path has positive probability of being completed before time t, so every entry of e^{tL_n} is positive. This matrix is stochastic. The Perron–Frobenius theorem therefore makes its eigenvalue one algebraically simple. A nontrivial Jordan chain of L_n at zero would exponentiate to a nontrivial Jordan chain of e^{tL_n} at one, which is impossible. Hence zero is algebraically simple. ■

Definition 6.2 (Spectral gap)**.** For $\omega > 0$ and $n \geq 2$, Lemma 6.1 makes zero a simple eigenvalue. Define

$$\gamma_n(\omega, \omega_0) = -\max\{\Re\lambda : \lambda \in \operatorname{spec} L_n(\omega, \omega_0),\ \lambda \neq 0\}.$$

Put $q = 2\pi/n$ and define the protected first-mode decay rate

$$\gamma_n^{\mathrm{hom}}(\omega) = -\Re\lambda_1^+ = 1 - \cos q + \omega - \Re\sqrt{\omega^2 - \sin^2 q}, \tag{6.1}$$

where the principal square root is used. For each fixed $\omega > 0$, the radicand is nonnegative for all sufficiently large n, and this rate then equals the homogeneous spectral gap. No such identity is asserted for small n, where the first-mode roots can be nonreal and the tumble mode can be slower.

Lemma 6.3 *(Fixed-rate regular Green estimate)***.** *Let $K \subset (0,\infty) \times [0,\infty)$ be compact and let $A > 0$. For $(\omega, \omega_0) \in K$, put*

$$p_n = \lambda_1^+, \qquad R_n = -\frac{\omega - r_1}{2r_1}, \qquad r_1 = \sqrt{\omega^2 - \sin^2(2\pi/n)}.$$

For all sufficiently large n, define the regular part on $|\lambda - p_n| \leq An^{-3}$ by

$$G_n^{\mathrm{reg}}(\lambda) = G_n(\lambda) - \frac{2R_n}{n(\lambda - p_n)}.$$

There is $C = C(K, A)$ such that throughout this disk

$$\left| G_n^{\mathrm{reg}}(\lambda) - \frac{1}{2(1+\omega)} \right| \leq \frac{C}{n}, \qquad |(G_n^{\mathrm{reg}})'(\lambda)| \leq Cn. \tag{6.2}$$

Proof. Write $q_k^* = 2\pi \min(k, n-k)/n$. Choose $\eta > 0$, uniformly for $(\omega, \omega_0) \in K$, so that $|\sin q| \leq \omega/2$ for $0 \leq q \leq \eta$. On this interval put

$$p^\pm(q) = \cos q - 1 - \omega \pm \sqrt{\omega^2 - \sin^2 q}, \qquad R^+(q) = -\frac{\omega - \sqrt{\omega^2 - \sin^2 q}}{2\sqrt{\omega^2 - \sin^2 q}}.$$

Taylor's theorem with remainder, applied uniformly on the compact ω-projection of K, gives constants $c, C > 0$ such that

$$cq^2 \le -p^+(q) \le Cq^2, \qquad |R^+(q)| \le Cq^2, \tag{6.3}$$
$$|p^+(q_k^*) - p^+(q_1^*)| \ge c(k^2-1)n^{-2} \qquad (2 \le k \le \eta n/(2\pi)). \tag{6.4}$$

For the last inequality, regard $p^+(q)$ as a function of q^2; its derivative at zero is $-(1+\omega)/(2\omega)$ and, after decreasing η, its modulus is bounded below by a positive constant. The fast root $p^-(q)$ stays a fixed distance from zero on $[0,\eta]$. For $\eta \le q \le \pi$, both roots stay a fixed distance from zero: the real-part dissipation of the homogeneous Fourier block can vanish only at $q = 0$ in the density direction, and the Heine–Borel theorem applied to the remaining compact parameter set supplies a uniform positive distance.

For an interior paired wave number with distinct roots, partial fractions give

$$\frac{\lambda + 1 - \cos q_k}{D_k(\lambda)} = \frac{R^+(q_k^*)}{\lambda - p^+(q_k^*)} + \frac{1 - R^+(q_k^*)}{\lambda - p^-(q_k^*)}.$$

Let S_n be the convex hull of zero and $\{|\lambda - p_n| \le An^{-3}\}$. Every point of S_n lies to the right of the second slow pole by at least $c(k^2-1)n^{-2}$ at index $k \ge 2$, after decreasing c and increasing n. Hence equations (6.3)–(6.4) bound the derivative of the regular slow-pole sum throughout S_n by

$$Cn \sum_{k=2}^{\infty} \frac{k^2}{(k^2-1)^2} \le C'n.$$

For $q_k^* \le \eta$, every fast-pole denominator stays uniformly away from S_n. For $q_k^* \ge \eta$, including a coalesced pair with $\omega = |\sin q_k|$, use the unreduced summand in (4.3). The compactness argument above gives $|D_k(\lambda)| \ge c > 0$ on S_n, uniformly in this range. The numerator, D_k, and D_k' are uniformly bounded there, so direct differentiation of $(\lambda + 1 - \cos q_k)/D_k(\lambda)$ is $O(1)$. After the $1/n$ normalization, each corresponding paired contribution to G_n is therefore $O(n^{-1})$, without dividing by the difference of the two roots. There are at most $n/2$ such pairs. The two endpoint contributions have derivatives $O(n^{-1})$ because their poles stay a fixed distance from S_n. Thus these terms have total derivative $O(1)$. This proves the derivative bound in (6.2).

At $\lambda = 0$, every unpaired nonzero-wave Fourier summand is exactly $1/[2(1+\omega)]$. The $q = 0$ term and the removed first-pole term each change the average by $O(n^{-1})$, using (6.3). Hence $G_n^{\mathrm{reg}}(0) = 1/[2(1+\omega)] + O(n^{-1})$. Finally $|p_n| = O(n^{-2})$, so integration of the derivative bound along the segment from zero to the displayed disk proves the first bound in (6.2). ∎

Lemma 6.4 *(Protected first mode and affected pole)*. *For every n, the protected sector contains λ_1^+ and $\Re\lambda_1^+ = -\gamma_n^{\mathrm{hom}}$. Whenever $|\sin(2\pi/n)| \le \omega$, this root is real and $\lambda_1^+ = -\gamma_n^{\mathrm{hom}}$. Let $K \subset (0,\infty) \times [0,\infty)$ be compact. Uniformly for $(\omega, \omega_0) \in K$ as $n \to \infty$, the affected root born from the same homogeneous pole is*

$$\lambda_{1,-} = \lambda_1^+ + \frac{\delta(1+\omega)}{\omega^2(1+\omega_0)} \frac{q^2}{n} + O(|\delta| n^{-4}). \tag{6.5}$$

Proof. The sector assertion is Theorem 4.4; the two equalities follow from Lemma 3.2 and Definition 6.2. Near λ_1^+, the two terms $k = 1, n-1$ in the full Fourier Green function have residue

$$R_1 = \frac{\lambda_1^+ + 1 - c_1}{\lambda_1^+ - \lambda_1^-} = -\frac{\omega - r_1}{2r_1}, \qquad r_1 = \sqrt{\omega^2 - \sin^2 q}.$$

Thus

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\mathrm{reg}}(\lambda).$$

If $\delta = 0$, the affected root equals the protected root and the formula follows. Suppose $\delta \neq 0$ and put

$$M(\lambda) = 1 + 2\delta G_n^{\mathrm{reg}}(\lambda), \qquad M_0 = \frac{1+\omega_0}{1+\omega}.$$

Lemma 6.3 gives, uniformly on every fixed $O(n^{-3})$ disk about λ_1^+,

$$M(\lambda) = M_0 + O(|\delta| n^{-1}), \qquad M'(\lambda) = O(|\delta| n).$$

The pole-cleared secular equation is

$$h_n(\lambda) := (\lambda - \lambda_1^+)M(\lambda) + \frac{4\delta R_1}{n} = 0.$$

Its affine comparison has the zero

$$\lambda_* - \lambda_1^+ = -\frac{4\delta R_1}{nM_0} = O(|\delta| n^{-3}).$$

On a circle of radius $C|\delta|n^{-4}$ about λ_*, the difference between h_n and $M_0(\lambda - \lambda_*)$ is at most $C_1\delta^2 n^{-4} + O(\delta^2 n^{-5})$. Since δ ranges over a fixed bounded set, choosing C larger than $C_1|\delta|/M_0$ makes the affine term strictly dominant. Rouché's theorem gives a unique root in this circle. Therefore

$$\lambda_{1,-} - \lambda_1^+ = \frac{2\delta(\omega - r_1)}{nr_1} \frac{1}{1 + \delta/(1+\omega)} + O(|\delta| n^{-4}),$$

where the constant is uniform on the compact parameter sets of the lemma. Finally $(\omega - r_1)/r_1 = q^2/(2\omega^2) + O(q^4)$, which proves (6.5). ∎

Lemma 6.5 *(Fixed-rate threshold Green bounds)***.** *Fix a compact $K \subset (0,\infty) \times [0,\infty)$, put $D = (1+\omega)/(2\omega)$ and $g_0 = 1/[2(1+\omega)]$, and let $p_n = \lambda_1^+$. The following estimates hold uniformly for $(\omega, \omega_0) \in K$.*

1. *For every $M < \infty$ there is C_M, independent of A and n, with the following property: for every fixed $A \geq 1$ and all $n \geq N(K, M, A)$, whenever $\lambda = \zeta/n^2$, $|\zeta| \leq M$, and*

$$\Re\lambda \geq \Re p_n - \pi^2 D n^{-2}, \qquad |\lambda - p_n| \geq A n^{-3}, \qquad A \geq 1,$$

$$|G_n(\lambda) - g_0| \leq \frac{C_M}{A} + \frac{C_M}{n}. \tag{6.6}$$

2. *For every fixed $C > 0$, if $\lambda_n \to 0$ satisfies $-Cn^{-2} \le \Re\lambda_n \le 0$ and $n^2|\lambda_n| \to \infty$, then*

$$G_n(\lambda_n) = g_0 + o(1). \tag{6.7}$$

Proof. Use the exact identities

$$x - 1 = \frac{\lambda(\lambda + 2\omega)}{2a}, \qquad \lambda + 1 - x = \frac{\lambda(\lambda + 2)}{2a}. \tag{6.8}$$

For $\lambda = \zeta/n^2$, choose either normalized local branch of $w_n = n \operatorname{arccosh} x$ satisfying $w_n/n \to 0$; the two such branches differ by sign. The products used below are invariant under that sign change. Taylor's theorem on $|\zeta| \le M$ and (6.8) give

$$w_n^2 = \frac{\zeta}{D} + O_{K,M}(n^{-2}), \tag{6.9}$$

$$\frac{\lambda + 1 - x}{2a\sinh(w_n/n)} = \frac{w_n}{4\omega(1+\omega)n} + O_{K,M}(|w_n|n^{-3}). \tag{6.10}$$

Here the quotients at $w_n = 0$ mean their analytic continuations. The first line in particular bounds $|w_n|$ uniformly. To verify the second line without dividing by a small w_n, divide the two identities in (6.8) and use $(\cosh t - 1)/\sinh t = \tanh(t/2)$. This gives the exact factorization

$$\frac{\lambda + 1 - x}{2a\sinh(w_n/n)} = \frac{\lambda + 2}{2a(\lambda + 2\omega)} \tanh\Big(\frac{w_n}{2n}\Big).$$

Uniformly on K and $|\zeta| \le M$, the prefactor on the right is $1/[2\omega(1+\omega)] + O_{K,M}(n^{-2})$, while Taylor's theorem gives $\tanh(w_n/(2n)) = w_n/(2n) + O_{K,M}(|w_n|^3 n^{-3})$. The uniform bound on $|w_n|$ proves (6.10), including at $w_n = 0$.

Since $z^n = e^{-w_n}$, the exact finite Green formula becomes

$$G_n(\zeta/n^2) = g_0 + \frac{w_n}{4\omega(1+\omega)n}\coth(w_n/2) + E_n, \tag{6.11}$$

where

$$|E_n| \le C_{K,M} n^{-2} + C_{K,M} n^{-3} |w_n \coth(w_n/2)|.$$

Let $q_\ell = 2\pi\ell/n$ for $1 \le \ell \le n/2$. Uniform Taylor expansion on the compact ω-projection of K gives

$$p^+(q_\ell) = -Dq_\ell^2 + O_K(n^{-4}) \qquad (\ell = 1, 2).$$

In particular,

$$p^+(q_2) - p_n = -12\pi^2 D n^{-2} + O_K(n^{-4}),$$
$$\Re p_n - \pi^2 D n^{-2} = -5\pi^2 D n^{-2} + O_K(n^{-4}).$$

Hence $p^+(q_2)$ lies strictly to the left of the stated half-plane for all sufficiently large n, uniformly on K. As in the proof of Lemma 6.3, $p^+(q)$ is a strictly decreasing function of q^2 on a fixed interval $0 \le q \le \eta$, while the fast branch there and both branches for $\eta \le q \le \pi$ stay a fixed distance to the left of zero. These facts exclude every remaining homogeneous pole from the half-plane. The zero density eigenvalue is removable in G_n, since its $q = 0$ summand reduces to $1/[n(\lambda + 2\omega)]$. The only nonzero pole in the stated scaled half-plane has $w_n = 2\pi\mathrm{i}$ up to sign and is exactly $\lambda = p_n$. The derivative of the analytic map $\zeta \mapsto w_n$ at this pole is bounded above and below away from zero. The complex inverse-function theorem, uniformly on the compact parameter set, therefore gives $|w_n - 2\pi\mathrm{i}| \geq cA/n$ outside $|\lambda - p_n| < An^{-3}$. On a fundamental strip,

$$|\coth(w/2)| \leq C\left(1 + \operatorname{dist}(w, 2\pi\mathrm{i}\mathbb{Z})^{-1}\right); \tag{6.12}$$

indeed, subtract the nearest $2\pi\mathrm{i}\ell$, use $\sinh(w/2) = w/2 + O(w^3)$ for $|w| \leq 1$, and use continuity on the remaining closed strip together with $\coth(w/2) \to \pm 1$ as $\Re w \to \pm\infty$. If $|w_n| \leq \eta$ for a sufficiently small fixed η, then $w_n \coth(w_n/2)$ is bounded by its analytic continuation at zero. Thus the correction term in (6.11) is $O_{K,M}(n^{-1})$ and $E_n = O_{K,M}(n^{-2})$ there. On the complementary region $|w_n| > \eta$, the uniform upper bound on $|w_n|$, the exclusion of the first pole, and (6.12) give

$$|\coth(w_n/2)| \leq C_{K,M}\left(1 + \frac{n}{A}\right).$$

The correction term is consequently bounded by $C_{K,M}/n + C_{K,M}/A$, and the new remainder bound gives $|E_n| \leq C_{K,M}n^{-2} + C_{K,M}/(An^2)$. This proves (6.6), including the removable point $w_n = 0$.

For item 2, first observe that λ_n is not a homogeneous pole for all sufficiently large n. Choose $\eta > 0$ as in the proof of Lemma 6.3. The fast branch on $[0, \eta]$ and both branches on $[\eta, \pi]$ stay uniformly away from zero. If a remaining pole were equal to λ_n, it would have the form $p^+(q_k^*)$; then (6.3) and $\Re\lambda_n \geq -Cn^{-2}$ would give $q_k^* = O(n^{-1})$ and hence $|\lambda_n| = O(n^{-2})$, contrary to $n^2|\lambda_n| \to \infty$. The real-part assumptions now imply $|\Im\lambda_n|/|\lambda_n| \to 1$. By the first identity in (6.8), $x - 1$ lies, for large n, in one of two closed sectors about the positive or negative imaginary axis. The expansion

$$\operatorname{arccosh}(1 + u) = \sqrt{2u}\,(1 + O(u))$$

on those sectors gives $\Re \operatorname{arccosh} x \geq c\sqrt{|\lambda_n|}$. Hence

$$|z(\lambda_n)|^n \leq \exp(-cn\sqrt{|\lambda_n|}) \longrightarrow 0.$$

Moreover, the two identities in (6.8) give $|(\lambda_n + 1 - x)/s| \leq C\sqrt{|\lambda_n|}$. Substitution in (5.12) now proves (6.7). ∎

Lemma 6.6 *(Right-edge root separation)***.** *Fix $\omega > 0$ and $\omega_0 \geq 0$, and put $D = (1+\omega)/(2\omega)$. There is N such that, for $n \geq N$, every nonzero eigenvalue other than the protected and affected roots born from $q = \pm 2\pi/n$ satisfies*

$$\Re\lambda \leq \Re\lambda_1^+ - \pi^2 D n^{-2}. \tag{6.13}$$

The constants can be chosen uniformly when (ω, ω_0) ranges over a compact set with ω bounded away from zero.

Proof. For the uniform assertion, fix a compact $K \subset (0,\infty)\times[0,\infty)$ containing the given parameter pair; the pointwise assertion follows from the same argument. All parameter-dependent objects below are evaluated at the current pair in K. Put

$$\Omega_n = \left\{\lambda : \Re\lambda \ge \Re\lambda_1^+ - \pi^2 D n^{-2}\right\}.$$

The protected roots are explicit. Their slow branch satisfies, uniformly for small real q,

$$\lambda^+(q) = -Dq^2 + O(q^4). \tag{6.14}$$

Choose $q_* > 0$ so that the remainder is at most $Dq^2/4$ for $|q| \le q_*$. The modes with $2\pi/n < |q| \le q_*$ are then to the left of Ω_n, while continuity puts both branches with $q_* \le |q| \le \pi$ a fixed distance into the left half-plane. The fast small-q branch is also separated from zero by a fixed amount. Thus, apart from zero, only the protected root born from $q = \pm 2\pi/n$ lies in Ω_n once n is large. We prove the same exclusion for the affected sector.

If $\delta = 0$, that sector is homogeneous and the conclusion follows from (6.14). Assume $\delta \ne 0$ and set

$$f_n(\lambda) = 1 + 2\delta G_n(\lambda), \qquad g_0 = \frac{1}{2(1+\omega)}, \qquad M_0 = 1 + 2\delta g_0 = \frac{1+\omega_0}{1+\omega} > 0. \tag{6.15}$$

When parameters vary below, Ω_n, f_n, g_0, and M_0 are evaluated at the current parameter pair. In a neighborhood of zero, every affected root is a zero of f_n. Indeed, the residue at every slow homogeneous pole is nonzero, so multiplication by $H^{(0)}_{n,-}$ cancels the pole of f_n but does not leave a zero there.

We first exclude, uniformly on K, sequences of affected zeros that do not tend to zero. Suppose otherwise. After discarding homogeneous terms, for which the assertion was already proved, there would be $n_j \to \infty$, parameters $(\omega^{(j)}, \omega_0^{(j)}) \in K$ with $\delta^{(j)} = \omega_0^{(j)} - \omega^{(j)} \ne 0$, and affected zeros $\lambda_j \in \Omega_{n_j}$, with the window evaluated at that pair, such that $|\lambda_j| \ge \epsilon > 0$. Gershgorin's theorem places these eigenvalues in a fixed disk and in the closed left half-plane. Pass to a subsequence such that

$$(\omega^{(j)}, \omega_0^{(j)}) \longrightarrow (\omega_*, \omega_{0,*}), \qquad \lambda_j \longrightarrow \lambda_* \ne 0.$$

Since the boundary of Ω_{n_j} tends to the imaginary axis, $\Re\lambda_* = 0$. For a bulk Bloch vector at wave number q, the real part of its quadratic form is the negative of $(1-\cos q)$ times its squared norm, together with the nonnegative tumbling form. Equality therefore forces $q = 0$ and equal orientation components, and the corresponding eigenvalue is zero. Thus Σ_{ω_*} meets the imaginary axis only at zero.

Choose a closed disk U about λ_* disjoint from zero, Σ_{ω_*}, and the point $a = 0$. By continuity of the bulk symbol and compactness, after shrinking U and increasing j, it is disjoint from $\Sigma_{\omega^{(j)}}$ and $a = 0$ as well. The local branches $z(\lambda; \omega^{(j)})$ can then be chosen jointly on U and satisfy $\sup_U |z(\lambda; \omega^{(j)})| \le \rho < 1$. Equation (5.13) gives the uniform estimate

$$\sup_{\lambda\in U} |G_{n_j}(\lambda; \omega^{(j)}) - g_\infty(\lambda; \omega^{(j)})| \le C_U \frac{\rho^{n_j}}{1-\rho^{n_j}} \longrightarrow 0.$$

For large j, the homogeneous odd factor is nonzero on U, so the affected polynomial equation there is the secular equation. It therefore passes to

$$1 + 2(\omega_{0,*} - \omega_*)g_\infty(\lambda_*;\omega_*) \;=\; 0.$$

If $\omega_{0,*} = \omega_*$ this equation is already impossible. Otherwise it produces an ℓ^2 eigenvector u of the limiting infinite defective generator with eigenvalue λ_*. This is impossible because

$$-\Re\langle u, L_\infty u\rangle \;=\; \frac{1}{2}\sum_{j,\sigma}|u^\sigma_{j+1} - u^\sigma_j|^2 + \sum_j \omega_j|u^+_j - u^-_j|^2. \tag{6.16}$$

Here (6.16) follows by expanding the inner product, shifting the two hopping sums, and pairing the two tumbling terms at each site. These operations are valid first for finitely supported vectors and then for all $u \in \ell^2$ by density and boundedness of L_∞. Equality in (6.16) forces an ℓ^2 vector to vanish. Consequently every sequence of affected zeros in Ω_n tends to zero, uniformly for parameters in K.

It remains to exhaust these affected zeros. On the $\zeta = n^2\lambda$ scale, the limiting first pole is $-4\pi^2 D(\omega)$, where $D(\omega) = (1+\omega)/(2\omega)$. Its range over the compact ω-projection of K is compact, and for each ω it stays uniformly separated from the corresponding limiting second pole. Choose a uniform neighborhood radius smaller than half this separation, and then choose M_* large enough to contain all the resulting first-pole neighborhoods. Let C_{M_*} be the uniform constant from item 1 of Lemma 6.5, and put

$$\Delta_K \;=\; \max_K |\omega_0 - \omega|, \qquad m_K \;=\; \min_K \frac{1+\omega_0}{1+\omega} \;>\; 0.$$

Choose A so large that $2\Delta_K C_{M_*}/A < m_K/4$. Only after these choices do we increase n. Suppose, to the contrary, that there were a sequence $n \to \infty$, parameters $(\omega_n, \omega_{0,n}) \in K$ with $\delta_n = \omega_{0,n} - \omega_n \neq 0$, and affected zeros $\lambda_n \in \Omega_n$ outside $|\lambda - \lambda_1^+| < An^{-3}$. Gershgorin's theorem gives $\Re\lambda_n \le 0$, and the preceding paragraph gives $\lambda_n \to 0$. The definition of Ω_n and (6.14) give $-Cn^{-2} \le \Re\lambda_n \le 0$. Pass to a subsequence according as $\zeta_n = n^2\lambda_n$ is bounded or unbounded. In the bounded case, pass once more so that the parameters converge and $\zeta_n \to \zeta_*$. If ζ_* is not the corresponding limiting first pole, (6.11)–(6.12) give $G_n(\lambda_n) \to g_0$. If it is the limiting first pole, then ζ_n lies in its chosen neighborhood for all sufficiently large n, and the first assertion of that lemma and the choice of A give $|G_n(\lambda_n) - g_0| \le C_{M_*}/A + o(1)$. In both alternatives, $|f_n(\lambda_n)| > m_K/2$ for large n, a contradiction. In the unbounded case, the second assertion of the lemma gives $G_n(\lambda_n) = g_0 + o(1)$ and the same contradiction. Hence, for all sufficiently large n, every affected zero in Ω_n lies in the first-pole disk, with a threshold uniform on K.

Finally, on the circle $|\lambda - \lambda_1^+| = An^{-3}$ write

$$G_n(\lambda) \;=\; \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\mathrm{reg}}(\lambda).$$

Lemma 6.3 and (6.15) give $1 + 2\delta G_n^{\mathrm{reg}} = M_0 + O(n^{-1})$. After multiplication by $\lambda - \lambda_1^+$, the secular equation becomes

$$(\lambda - \lambda_1^+)f_n(\lambda) \;=\; M_0(\lambda - \lambda_1^+) + \frac{4\delta R_1}{n} + O(n^{-1}|\lambda - \lambda_1^+|). \tag{6.17}$$

Here $R_1 = O(n^{-2})$. Enlarge A, if necessary, so that the zero of the first two terms lies strictly inside the circle. On the circle the last term is $O(n^{-4})$, whereas the first two terms have modulus bounded below by $c_A n^{-3}$. Rouché's theorem therefore counts exactly one zero in the disk. It is the affected root of Lemma 6.4. Thus no other affected root occurs in Ω_n, which proves the assertion.

All estimates above are uniform on the stated compact parameter sets: D, δ, and the generator norms are bounded there, while M_0 has a positive minimum. ∎

Theorem 6.7 *(Fixed-rate gap asymptotics). Fix $\omega > 0$ and $\omega_0 \geq 0$.*

1. *If $\omega_0 \leq \omega$, then for every sufficiently large n,*

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\mathrm{hom}}(\omega). \tag{6.18}$$

2. *If $\omega_0 > \omega$, then*

$$\gamma_n(\omega, \omega_0) = \gamma_n^{\mathrm{hom}}(\omega) - \frac{4\pi^2(\omega_0 - \omega)(1+\omega)}{\omega^2(1+\omega_0)} n^{-3} + O(|\omega_0 - \omega| n^{-4}). \tag{6.19}$$

3. *In both cases,*

$$\begin{aligned} \gamma_n^{\mathrm{hom}}(\omega) = {} & \frac{2\pi^2(1+\omega)}{\omega} n^{-2} \\ & - 16\pi^4 \left(\frac{1}{24} + \frac{1}{6\omega} - \frac{1}{8\omega^3} \right) n^{-4} + O(n^{-6}). \end{aligned} \tag{6.20}$$

The errors are uniform when (ω, ω_0) ranges over a compact set with ω bounded away from zero. The root selection is uniform through $\omega_0 = \omega$, where the two parity roots coincide.

Proof. Taylor expansion of (6.1) gives (6.20). Lemma 6.4 shows that, uniformly across $\delta = 0$, the affected first root moves toward zero precisely when $\delta > 0$. Lemma 6.6 shows that no other extended, band-edge, or localized root can compete with this first pair. Therefore the protected root wins when $\delta \leq 0$, proving (6.18), and the affected root wins when $\delta > 0$. Substitution of $q^2 = 4\pi^2 n^{-2}$ in (6.5) then gives (6.19). ∎

Corollary 6.8 *(Fixed ratio, vanishing ratio, and a slow-site scaling). Let the bulk rate $\omega > 0$ be fixed.*

1. *If $\omega_0/\omega \to \rho \in (0, \infty)$, equations (6.18)–(6.20) apply. For $\rho < 1$ the protected branch is selected; for $\rho > 1$ the affected branch and its n^{-3} term are selected. At the boundary $\rho = 1$, the protected root is selected whenever $\omega_0 \leq \omega$ and the affected root whenever $\omega_0 > \omega$, with the expansion uniform across equality.*
2. *If $\omega_0/\omega \to 0$, then (6.18) holds for every sufficiently large n.*
3. *In particular, if $n\omega_0 \to \beta$ with $\beta \geq 0$, then (6.18) still holds. A single slow tumbling site therefore has no separate $\omega_0 \asymp n^{-1}$ gap crossover when the bulk rate is fixed.*

Proof. In items 2 and 3 one has $\omega_0 < \omega$ for every sufficiently large n, so Theorem 6.7 applies. In item 1 the convergence of ω_0/ω places the n-dependent pairs (ω, ω_0) in one compact parameter set. The compact-uniform statement in Theorem 6.7 therefore permits this n-dependence, and its sign test gives the asserted branch selection, including crossings through equality. ∎

Definition 6.9 (Critical homogeneous poles and residues)**.** Let $\alpha > 0$, set $\omega = \alpha/n$ and $k = 2\pi$, and, for an integer ℓ, put

$$\begin{aligned} p^{\pm}_{\ell,n} &= \cos(\ell k/n) - 1 - \frac{\alpha}{n} \pm \sqrt{\frac{\alpha^2}{n^2} - \sin^2(\ell k/n)}, \\ R^{\pm}_{\ell,n} &= \frac{p^{\pm}_{\ell,n} + 1 - \cos(\ell k/n)}{p^{\pm}_{\ell,n} - p^{\mp}_{\ell,n}}. \end{aligned} \tag{6.21}$$

The square root is chosen nonnegative when its argument is nonnegative and with nonnegative imaginary part when its argument is negative. When the two poles are distinct and $1 \le \ell < n/2$, the paired waves $q = \pm\ell k/n$ contribute $2R^{\pm}_{\ell,n}/\{n(\lambda - p^{\pm}_{\ell,n})\}$ for the indicated sign. The unpaired endpoint contributions are instead

$$\frac{1}{n(\lambda + 2\alpha/n)}, \qquad \frac{\varepsilon_n}{n(\lambda + 2 + 2\alpha/n)},$$

at $q = 0$ and $q = \pi$, respectively.

Lemma 6.10 *(Critical Green limit and pole remainders)***.** *Set $\omega = \alpha/n$, where $\alpha > 0$ is fixed, and put $k = 2\pi$. For $\nu \in \mathbb{C}$, define the meromorphic function*

$$\mathcal{G}_\alpha(\nu) = \frac{1}{2}\left[1 + \frac{\nu}{\kappa}\coth\left(\frac{\kappa}{2}\right)\right], \qquad \kappa^2 = \nu(\nu + 2\alpha), \tag{6.22}$$

where the right-hand side is independent of the choice of square root and is interpreted by meromorphic continuation at $\kappa = 0$. The decompositions in items 2–4 are identities of meromorphic functions. At each removed pole, the difference defining the regular part is evaluated by its unique holomorphic extension. Then:

1. *$G_n(\nu/n) \to \mathcal{G}_\alpha(\nu)$ locally uniformly on every compact set containing no pole of (6.22).*
2. *Fix $\ell \ge 1$ with $\alpha \ne \ell k$, and let $p^{\pm}_{\ell,n}$ and $R^{\pm}_{\ell,n}$ be given by (6.21). There are $\eta, C > 0$ such that*
$$\left| G_n(\lambda) - \frac{2R^{\pm}_{\ell,n}}{n(\lambda - p^{\pm}_{\ell,n})} \right| \le C \tag{6.23}$$
whenever $|n\lambda - np^{\pm}_{\ell,n}| \le \eta$, for all sufficiently large n.
3. *If $\alpha = k$, put*
$$\lambda_c = \cos(k/n) - 1 - k/n, \qquad r_n = \sqrt{(k/n)^2 - \sin^2(k/n)}.$$
There are $\eta, C > 0$ and analytic functions H_n on $|n(\lambda - \lambda_c)| \le \eta$ such that
$$G_n(\lambda) = \frac{2(\lambda - \lambda_c - k/n)}{n\{(\lambda - \lambda_c)^2 - r_n^2\}} + H_n(\lambda), \qquad |H_n(\lambda)| \le C. \tag{6.24}$$

4. *There are $\eta, C > 0$ and analytic functions K_n on $|n\lambda + 2\alpha| \le \eta$ such that*

$$G_n(\lambda) = \frac{1}{n(\lambda + 2\alpha/n)} + K_n(\lambda), \qquad |K_n(\lambda)| \le C. \tag{6.25}$$

Proof. Let $w_n(\nu) = n \operatorname{arccosh} x(\nu/n)$ on a local branch satisfying $w_n/n \to 0$; such normalized branches differ only by sign. The exact identities (6.8) give

$$\cosh(w_n/n) = 1 + \frac{\nu(\nu+2\alpha)}{2n^2\{1 + (\nu+\alpha)/n\}}, \tag{6.26}$$

$$\frac{\lambda + 1 - x}{s} = \frac{\nu(2 + \nu/n)}{2n\{1 + (\nu+\alpha)/n\}\sinh(w_n/n)}, \qquad \lambda = \nu/n. \tag{6.27}$$

On a compact ν-set, Taylor's theorem applied to (6.26) yields $w_n^2 = \nu(\nu+2\alpha) + O(n^{-1})$ uniformly. Substitution of (6.27) into the exact formula (5.12), with $(1+z^n)/(1-z^n) = \coth(w_n/2)$, gives the pointwise limit in item 1. The resulting expression is even and meromorphic in w_n, hence meromorphic in w_n^2, with its value at $w_n = 0$ taken by continuation. On a compact set avoiding the limiting poles, the denominators in this even expression stay uniformly separated from zero. Uniform convergence of w_n^2, together with the explicit coefficients in (6.27), therefore gives uniform convergence there. A general branch change has the form $w_n \mapsto \pm w_n + 2\pi i n m$; the sign changes in $\sinh(w_n/n)$ and $\coth(w_n/2)$ cancel, and the additive term is a period of both factors. Thus the conclusion agrees on overlapping branch neighborhoods and is independent of the normalized branch.

For item 2, the limiting poles are

$$\nu_\ell^\pm = -\alpha \pm \sqrt{\alpha^2 - \ell^2 k^2}.$$

The hypothesis $\alpha \neq \ell k$ makes them simple. Choose a closed disk U about one of them that contains no other limiting pole. For large n, $G_n(\nu/n)$ has exactly the corresponding pole $np_{\ell,n}^\pm$ in U, with principal part $2R_{\ell,n}^\pm/(\nu - np_{\ell,n}^\pm)$. Item 1 bounds both the Green function and this principal part on ∂U. Their difference is removable at the corresponding pole and extends analytically throughout U, so the maximum-modulus theorem bounds it on a smaller concentric disk. Replacing ν by $n\lambda$ gives (6.23).

When $\alpha = k$, the two first poles lie at $\lambda_c \pm r_n$. Adding their two summands in (4.3) gives exactly the displayed rational term in (6.24). Choose a fixed disk about $-k$ in the ν-plane that contains no other limiting pole. On its boundary, item 1 bounds $G_n(\nu/n)$ and direct inspection bounds the grouped term. Their difference has removable singularities at both first poles and extends analytically inside. The maximum-modulus theorem gives the asserted uniform bound on a smaller disk.

Finally, the first summand of (4.3) is exactly $1/\{n(\lambda + 2\alpha/n)\}$. The point -2α is distinct from all the limiting poles with $\ell \ge 1$. Choose a fixed disk about it in the ν-plane containing no such pole. For large n, the corresponding scaled disk contains no finite-ring pole except the $q = 0$ pole. On the boundary, item 1 bounds $G_n(\nu/n)$ and the displayed summand; their difference has a removable singularity at the endpoint pole and extends analytically to the disk. The maximum-modulus theorem on a smaller disk proves (6.25). ∎

Lemma 6.11 *(Critical pole-distance conversion)***.** *Fix $\alpha > 0$ and $R < \infty$, put $\omega = \alpha/n$ and $k = 2\pi$, and let w satisfy $\cosh(w/n) = x(\lambda)$ in either right-edge window below. Define the branch-invariant distances*

$$d_{0,n}(w) = \operatorname{dist}(w, 2\pi \mathrm{i} n\mathbb{Z}), \qquad d_{*,n}(w) = \operatorname{dist}\big(w, 2\pi\mathrm{i}(\mathbb{Z}\setminus n\mathbb{Z})\big).$$

Thus $d_{0,n}$ measures distance to the zero-wave lattice points. In the right-edge windows below, the nonremovable tumble pole $\lambda = -2\omega$ lies outside the window, so only the removable $\lambda = 0$ branch is relevant there. The distance $d_{*,n}$ measures distance to every nonzero-wave pole. The following estimates hold for $|\lambda| \le R$.

1. *Suppose $\alpha \le k$. For an integer $L \ge 1$, set*

$$B = \frac{k^2}{4}\{L^2 + (L+1)^2\}, \qquad \Omega_{B,n} = \{\Re\lambda \ge -\alpha/n - B/n^2\}.$$

   Delete disks of radius An^{-2} about the simple homogeneous poles in $\Omega_{B,n}$. When $\alpha = k$, replace the two first-pole disks by the disk of radius $An^{-3/2}$ about their midpoint. There are $C = C(\alpha, R)$ and $C_B = C_B(\alpha, B, R)$ such that, for every fixed $A \ge 1$ and all sufficiently large n,

$$d_{*,n}(w)^{-1} \le C_B\Big(1 + \frac{n}{A}\Big) + C\frac{n}{\sqrt{B}} \tag{6.28}$$

   on the remaining part of $\Omega_{B,n}$.

2. *Suppose $\alpha > k$. Put*

$$d_1 = \alpha - \sqrt{\alpha^2 - k^2}, \qquad d_2 = \alpha - \Re\sqrt{\alpha^2 - 4k^2},$$

   fix $d \in (d_1, d_2)$, and set $\Omega_{d,n} = \{\Re\lambda \ge -d/n\}$. After deleting the disk of radius An^{-2} about the slow first pole, there is $C_d = C_d(\alpha, d, R)$ such that

$$d_{*,n}(w)^{-1} \le C_d\Big(1 + \frac{n}{A}\Big) \tag{6.29}$$

   throughout the remaining part of $\Omega_{d,n}$, for every fixed $A \ge 1$ and all sufficiently large n.

The thresholds for n may depend on the displayed fixed parameters and on A, but the constants in (6.28)– (6.29) do not depend on A or n.

Proof. A branch change has the form $w \mapsto \pm w + 2\pi\mathrm{i}nm$. Both pole sets in the definition are invariant under this transformation, so both distances are branch independent. It is enough to consider $d_{*,n}(w) < 1$. Choose the nearest nonzero-wave representative and use the branch symmetries to write it as $2\pi\mathrm{i}\ell$, where $1 \le \ell \le \lfloor n/2 \rfloor$, and put

$$q = \frac{\ell k}{n}, \qquad \theta = w - 2\pi\mathrm{i}\ell.$$

The exact factorization

$$2a\{\cosh(w/n) - \cos(\ell k/n)\} = D_\ell(\lambda) = (\lambda - p^+_{\ell,n})(\lambda - p^-_{\ell,n}) \tag{6.30}$$

shows that the two zeros over this representative are precisely the two homogeneous poles of wave number q. Taylor's theorem, $|a| \le C_R$, and $|\theta| < 1$ give

$$|D_\ell(\lambda)| \le C_R \left(\frac{|\sin q|}{n} |\theta| + \frac{|\theta|^2}{n^2} \right). \tag{6.31}$$

Suppose first that $\alpha \le k$. The poles with indices $1 \le \ell \le L$ lie in $\Omega_{B,n}$ for all sufficiently large n. Except for $(\alpha, \ell) = (k, 1)$, they are simple. On the scaled variable $\nu = n\lambda$, the limiting relation is $w^2 = \nu(\nu + 2\alpha)$, and at a simple limiting pole

$$\frac{dw}{d\nu} = \frac{\nu + \alpha}{w} \ne 0.$$

For each of these fixed indices, (6.31) first gives $|D_\ell(\lambda)| \le C_B n^{-2}$. The factorization in (6.30) therefore gives

$$\min_{\sigma \in \{+,-\}} |\lambda - p^\sigma_{\ell,n}|^2 \le |(\lambda - p^+_{\ell,n})(\lambda - p^-_{\ell,n})| \le C_B n^{-2}.$$

Both poles are $O(n^{-1})$, so $|\lambda| \le C'_B n^{-1}$ and hence $\nu = n\lambda$ lies in a fixed compact set. To make the finite-n uniformity explicit, write the local branch as $w = w_n(\nu)$. Differentiating the exact relation $\cosh(w_n/n) = x(\nu/n)$ gives

$$\frac{\partial w_n}{\partial \nu} = \frac{x'(\nu/n)}{\sinh(w_n/n)}.$$

The identity

$$x(\lambda) - 1 = \frac{\lambda(\lambda + 2\omega)}{2(1 + \omega + \lambda)}$$

then gives, uniformly on each of these fixed scaled neighborhoods,

$$\frac{\partial w_n}{\partial \nu} = \frac{\nu + \alpha}{w_n} + O_B(n^{-1}).$$

After shrinking these finitely many neighborhoods, choose positive constants c_B, C_B, and r_B, independent of n, with the following property. For each corresponding finite pole, set

$$\nu^\pm_{\ell,n} = n p^\pm_{\ell,n},$$

One may choose the local branch so that $w_n(\nu^\pm_{\ell,n}) = 2\pi \mathrm{i}\ell$ and

$$\left| w'_n(\nu^\pm_{\ell,n}) \right| \ge 2c_B, \qquad \sup_{|\nu - \nu^\pm_{\ell,n}| \le r_B} |w''_n(\nu)| \le C_B.$$

The first bound follows from the displayed derivative and the simple limiting pole. The second follows from Cauchy's estimate on slightly larger fixed neighborhoods, using uniform analytic convergence there. Decrease r_B so that $C_B r_B/2 \le c_B$. Taylor's formula then gives

$$|w_n(\nu) - 2\pi \mathrm{i}\ell| \ge c_B |\nu - \nu^\pm_{\ell,n}| \qquad (|\nu - \nu^\pm_{\ell,n}| \le r_B).$$

Outside the deleted An^{-2} disk this is at least $c_B A/n$. On the complement of those fixed scaled neighborhoods inside the compact ν-set just obtained, the limiting branch has positive distance from the corresponding representative $2\pi \mathrm{i}\ell$. Uniform convergence and compactness therefore give a positive lower bound for the finite-n distance. The reciprocal contribution of all these finitely many indices is thus at most $C_B(1 + n/A)$.

At $\alpha = k$ and $\ell = 1$, write $\Delta = \lambda - \lambda_c$. The two first poles are $\lambda_c \pm r_n$, with $r_n = O(n^{-2})$. Outside the grouped disk,

$$|D_1(\lambda)| = |\Delta^2 - r_n^2| \geq cA^2 n^{-3}.$$

Since $|\sin(k/n)| \asymp n^{-1}$, (6.31) implies $|\theta| \geq cA^2/n \geq cA/n$. This supplies the same reciprocal bound as a simple pole without separating the coalescing pair.

It remains to treat $\ell \geq L+1$. Choose $q_0 \in (0, \pi/2)$. When $q \leq q_0$, the two poles are conjugate for large n; write them as $h_{\ell,n} \pm \mathrm{i}t_{\ell,n}$. Every point of $\Omega_{B,n}$ has horizontal separation at least

$$u_{\ell,n} = 1 - \cos(\ell k/n) - B/n^2$$

from their common real part. The midpoint definition of B and $1 - \cos q = q^2/2 + O(q^4)$ give

$$u_{\ell,n} \geq u_{L+1,n} \geq c\sqrt{B}\, n^{-2}, \qquad t_{\ell,n} \geq cq.$$

For $u \geq 0$ and real y, t,

$$|(u + \mathrm{i}(y - t))(u + \mathrm{i}(y + t))| \geq 2u|t|, \tag{6.32}$$

because the difference of the squares of the two sides is $(u^2 + y^2 - t^2)^2$. Hence $|D_\ell(\lambda)| \geq cu_{\ell,n}q$. Since $q \geq 2k/n$ and $|\theta| < 1$, the right side of (6.31) is at most $Cq|\theta|/n$. Therefore

$$|\theta| \geq cnu_{\ell,n} \geq c\sqrt{B}/n.$$

When $q \in [q_0, \pi]$, both poles have real part at most $-c(q_0) < 0$, while the boundary of $\Omega_{B,n}$ tends to zero. Thus $|D_\ell(\lambda)| \geq c(q_0)^2$, which is incompatible with (6.31) and $|\theta| < 1$ for large n. This proves (6.28).

Now suppose $\alpha > k$. The choice $d_1 < d < d_2$ places only the slow first pole in $\Omega_{d,n}$. The simple-pole inverse-function argument gives $|\theta| \geq cA/n$ near that pole after its disk is deleted. The fast first pole and the complement of a fixed scaled neighborhood of the slow pole have a positive scaled separation from the window, so they give $|\theta| \geq c$.

Choose L_0 with $L_0 k > 2\alpha$. For each fixed $2 \leq \ell \leq L_0$, both pole factors have horizontal distance at least c/n from $\Omega_{d,n}$; hence $|D_\ell(\lambda)| \geq cn^{-2}$. Equation (6.31) gives $|D_\ell(\lambda)| \leq C|\theta|n^{-2}$ when $|\theta| < 1$, so $|\theta| \geq c$. This also covers a coalescence $\alpha = \ell k$.

For $\ell > L_0$ and $q \leq q_0$, the poles are conjugate, $t_{\ell,n} \geq cq$, and their horizontal separation from the window is at least

$$u_{\ell,n} = \frac{\alpha - d}{n} + 1 - \cos q.$$

Equations (6.32) and (6.31) give

$$|\theta| \geq cnu_{\ell,n} \geq c(\alpha - d).$$

The fixed separation for $q \in [q_0, \pi]$ again rules out $|\theta| < 1$. Combining the slow-pole estimate with these fixed lower bounds proves (6.29). ∎

Lemma 6.12 *(Critical off-pole Green estimate)***.** *Fix $\alpha > 0$ and $R < \infty$, put $\omega = \alpha/n$ and $k = 2\pi$, and restrict λ to $|\lambda| \le R$. The following estimates hold.*

1. *Suppose $\alpha \le k$. For an integer $L \ge 1$, set*

$$B = \frac{k^2}{4}\{L^2 + (L+1)^2\}, \qquad \Omega_{B,n} = \{\Re\lambda \ge -\alpha/n - B/n^2\}.$$

Delete the disks of radius An^{-2} about all simple homogeneous poles in $\Omega_{B,n}$. If $\alpha = k$, replace the two disks at the first poles by the disk of radius $An^{-3/2}$ about their midpoint. There are constants $C = C(\alpha, R)$ and $C_B = C_B(\alpha, B, R)$, independent of A and n, such that, for every fixed $A \ge 1$, the remaining set satisfies the following bound whenever $n \ge N(\alpha, B, R, A)$:

$$|G_n(\lambda)| \le C_B\left(1 + \frac{n}{A}\right) + C\frac{n}{\sqrt{B}}. \tag{6.33}$$

2. *Suppose $\alpha > k$. Define*

$$d_1 = \alpha - \sqrt{\alpha^2 - k^2}, \qquad d_2 = \alpha - \Re\sqrt{\alpha^2 - 4k^2},$$

choose $d \in (d_1, d_2)$, and put $\Omega_{d,n} = \{\Re\lambda \ge -d/n\}$. After deleting the disk of radius An^{-2} about the slow first pole, one has

$$|G_n(\lambda)| \le C_d\left(1 + \frac{n}{A}\right) \tag{6.34}$$

throughout $\Omega_{d,n} \cap \{|\lambda| \le R\}$. Here $C_d = C_d(\alpha, d, R)$ is independent of A and n, and, for every fixed $A \ge 1$, the estimate holds for $n \ge N(\alpha, d, R, A)$.

Proof. Write $w = n \operatorname{arccosh} x$ and $\Theta = (\lambda + 1 - x)/s$. Besides (5.12), we use the exact identity

$$\Theta^2 = \frac{\lambda(\lambda+2)}{(\lambda+2\omega)(\lambda+2+2\omega)}. \tag{6.35}$$

It follows by squaring the second identity in (6.8) and using

$$x + 1 = \frac{(\lambda+2)(\lambda+2+2\omega)}{2a}.$$

A branch change replaces w by $\pm w + 2\pi i n m$. Under the sign change both $s = \sinh(w/n)$ and $\coth(w/2)$ change sign, while the additive term is a period. Thus $\Theta \coth(w/2)$ is unchanged, as are the distances in Lemma 6.11.

The right-edge windows stay a fixed multiple of n^{-1} to the right of the tumble pole -2ω and a fixed distance to the right of $-2 - 2\omega$. Consequently $|a|$ is bounded below and (6.35) gives $|\Theta| \le C$ there. Fix a small $\eta > 0$. If $d_{0,n}(w) \le \eta$, a branch change puts $|w| \le \eta$. Taking the quotient of the two identities in (6.8) where it is defined, extending through removable points by analytic continuation, and using $(\cosh t - 1)/\sinh t \;=\; \tanh(t/2)$ gives the exact factorization

$$\Theta \coth(w/2) \;=\; \frac{\lambda+2}{\lambda+2\omega}\tanh\Big(\frac{w}{2n}\Big)\coth(w/2). \tag{6.36}$$

In the subcritical window,

$$\Re(\lambda+2\omega) \;\geq\; \frac{\alpha}{n} - \frac{B}{n^2} \;\geq\; \frac{\alpha}{2n}$$

for all sufficiently large n. In the supercritical window,

$$\Re(\lambda+2\omega) \;\geq\; \frac{2\alpha-d}{n},$$

where $2\alpha - d \;>\; 0$ because $d \;<\; d_2 \;\leq\; \alpha$. Since also $|\lambda+2\omega| \;\geq\; |\Im\lambda|$, the two cases give

$$|\lambda+2\omega| \;\geq\; c(n^{-1} + |\Im\lambda|). \tag{6.37}$$

Since $|w| \;\leq\; \eta$, the first identity in (6.8) and the upper bound on $|a|$ imply $|\lambda(\lambda+2\omega)| \;\leq\; Cn^{-2}$, and hence $|\lambda| \;\leq\; Cn^{-1}$. Indeed, if $|\lambda| \;\leq\; 4\omega \;=\; 4\alpha/n$, this is immediate; otherwise $|\lambda+2\omega| \;\geq\; |\lambda| - 2\omega \;\geq\; |\lambda|/2$, and therefore $|\lambda|^2 \;\leq\; 2Cn^{-2}$. The lower bound on $|a|$ and the same identity then give

$$|w|^2 \;\leq\; Cn^2|\lambda|\,|\lambda+2\omega| \;\leq\; Cn|\lambda+2\omega|.$$

Finally, Taylor's theorem gives, uniformly for $n \;\geq\; 1$ and $|w| \;\leq\; \eta$,

$$n\tanh\Big(\frac{w}{2n}\Big)\coth(w/2) \;=\; 1 + \mathcal{E}_n(w), \qquad |\mathcal{E}_n(w)| \;\leq\; C|w|^2.$$

Moreover, the preceding bound on $|w|^2$ gives

$$\left|\frac{\lambda+2}{n(\lambda+2\omega)}\mathcal{E}_n(w)\right| \;\leq\; \frac{C_R|w|^2}{n|\lambda+2\omega|} \;\leq\; C_R.$$

Substitution in (6.36) therefore gives

$$\Theta\coth(w/2) \;=\; \frac{\lambda+2}{n(\lambda+2\omega)} + O(1), \tag{6.38}$$

uniformly in either window. Equation (6.37) shows that the leading term, and hence the entire expression, is bounded.

If $d_{0,n}(w) \;>\; \eta$, then

$$\operatorname{dist}(w, 2\pi\mathrm{i}\mathbb{Z}) \;=\; \min\{d_{0,n}(w), d_{*,n}(w)\}.$$

The finite Green formula and (6.12) therefore give, throughout this complementary region,

$$|G_n(\lambda)| \;\leq\; C\left\{1 + d_{*,n}(w)^{-1}\right\}. \tag{6.39}$$

Equation (6.38) bounds the first region. On the second region, combine (6.39) with item 1 or item 2 of Lemma 6.11. This gives respectively (6.33) and (6.34), with exactly the stated dependence of the constants. ■

Lemma 6.13 *(Critical pole-neighborhood root counts). Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, put $k = 2\pi$, and write $b = \beta - \alpha$.*

1. *Fix $\ell \geq 1$ with $\alpha \neq \ell k$. There is $A_\ell > 0$ such that, for each sign, every fixed $A \geq A_\ell$, and all sufficiently large n, the disk $|\lambda - p^\pm_{\ell,n}| < An^{-2}$ contains exactly one zero $\widehat{p}^\pm_{\ell,n}$ of $Q_{n,-}$, counted with algebraic multiplicity, and*
$$\widehat{p}^\pm_{\ell,n} = p^\pm_{\ell,n} - 4bR^\pm_{\ell,n} n^{-2} + O(n^{-3}). \tag{6.40}$$
The threshold in n may depend on A.
2. *If $\alpha = k$, put*
$$\lambda_c = \cos(k/n) - 1 - k/n, \qquad r_n = \sqrt{(k/n)^2 - \sin^2(k/n)}.$$
There is $A_* > 0$ such that, for every fixed $A \geq A_*$ and all sufficiently large n, the disk $|\lambda - \lambda_c| < An^{-3/2}$ contains exactly two zeros of $Q_{n,-}$, counted with algebraic multiplicity. The same two zeros occur for every such fixed enlargement A.

Proof. If $b = 0$, then $Q_{n,-} = H^{(0)}_{n,-}$. A simple-pole disk contains exactly its homogeneous root, so item 1 holds with zero shift. At $\alpha = k$, the two first homogeneous roots are $\lambda_c \pm r_n$, where $r_n = O(n^{-2})$; both lie in every fixed $An^{-3/2}$ disk for all sufficiently large n, while all other homogeneous roots are separated on the n^{-1} scale. Thus item 2 also holds. Assume below that $b \neq 0$, so $\delta = b/n$.

At a simple pole, Lemma 6.10 gives
$$G_n(\lambda) = \frac{2R^\pm_{\ell,n}}{n(\lambda - p^\pm_{\ell,n})} + G^{\text{reg}}_{\ell,n}(\lambda), \qquad G^{\text{reg}}_{\ell,n} = O(1)$$
on a fixed scaled neighborhood. Multiplication of the secular equation by $\lambda - p^\pm_{\ell,n}$ gives
$$(\lambda - p^\pm_{\ell,n})(1 + 2\delta G_n) = (\lambda - p^\pm_{\ell,n})\left(1 + \frac{2b}{n} G^{\text{reg}}_{\ell,n}\right) + \frac{4bR^\pm_{\ell,n}}{n^2}. \tag{6.41}$$
Compare this expression with
$$h_0(\lambda) = \lambda - p^\pm_{\ell,n} + 4bR^\pm_{\ell,n} n^{-2}.$$
If $\sigma_\ell = \sqrt{\alpha^2 - \ell^2 k^2}$ with the branch inherited from Definition 6.9, then
$$R^\pm_{\ell,n} \longrightarrow \frac{1}{2} \mp \frac{\alpha}{2\sigma_\ell} \neq 0;$$
the inequality follows from $\ell \geq 1$. Thus the residues remain bounded and bounded away from zero. Choose A_ℓ so that the zero of h_0 lies a fixed relative distance inside the disk. On its boundary, the difference from h_0 is $O(n^{-3})$, whereas $|h_0| \geq c_A n^{-2}$. Rouché's theorem gives exactly one pole-cleared zero. At that zero, (6.41) first gives $\lambda - p^\pm_{\ell,n} = O(n^{-2})$ and then, after substitution in its regular-part term, gives the expansion (6.40).

On this disk write $H^{(0)}_{n,-} = (\lambda - p^{\pm}_{\ell,n})J_{\ell,n}$, where $J_{\ell,n}$ is nonvanishing. The residue $R^{\pm}_{\ell,n}$ is nonzero, and the right side of (6.41) is nonzero at the homogeneous pole. Thus that pole is canceled rather than retained, and the Rouché zero is exactly one zero of $Q_{n,-}$ with its algebraic multiplicity. This proves item 1.

Now let $\alpha = k$ and put $\Delta = \lambda - \lambda_c$. By Lemma 6.10,

$$(\Delta^2 - r_n^2)(1 + 2\delta G_n) = (\Delta^2 - r_n^2)\left(1 + \frac{2b}{n}H_n\right) + \frac{4b}{n^2}(\Delta - k/n), \tag{6.42}$$

where $H_n = O(1)$ throughout the grouped disk. Compare this with

$$P_{0,n}(\Delta) = \Delta^2 - r_n^2 + \frac{4b}{n^2}(\Delta - k/n).$$

With $\Delta = n^{-3/2}Y$, one has locally uniformly in Y

$$n^3 P_{0,n}(n^{-3/2}Y) = Y^2 - 4bk + O(n^{-1/2}).$$

Choose $A_* > 2\sqrt{|b|k}$. Rouché's theorem applied to this scaled quadratic shows that both zeros of $P_{0,n}$ lie a fixed relative distance inside $|\Delta| < A_* n^{-3/2}$. On the boundary of every fixed enlargement, the difference in (6.42) is $O(A^2 n^{-4})$, whereas $|P_{0,n}| \geq c_A n^{-3}$. Rouché's theorem gives exactly two pole-cleared zeros. At the homogeneous poles $\Delta = \pm r_n$, the right side of (6.42) equals

$$\frac{4b}{n^2}(\pm r_n - k/n),$$

which is nonzero for all sufficiently large n, because $r_n = k^2/(\sqrt{3}\,n^2) + O(n^{-4})$. The factor $\Delta^2 - r_n^2$ is the complete homogeneous factor on the disk and the remaining factor of $H^{(0)}_{n,-}$ is nonvanishing. Hence the two Rouché zeros are exactly the two zeros of $Q_{n,-}$ there. Applying the count once at A_* and once at any fixed $A \geq A_*$ shows that the annular difference contains no zero, proving item 2. ∎

Lemma 6.14 *(Critical zero-wave affected root)***.** *Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, and put $b = \beta - \alpha$. For all sufficiently large n, the affected polynomial $Q_{n,-}$ has exactly one zero in a fixed $O(n^{-3})$ neighborhood of $-2\alpha/n - 2b/n^2$. Counted with algebraic multiplicity, it is*

$$\lambda_{0,-} = -\frac{2\alpha}{n} - \frac{2b}{n^2} + O(n^{-3}). \tag{6.43}$$

If $b \neq 0$, the homogeneous pole $-2\alpha/n$ is canceled; if $b = 0$, the displayed zero is that homogeneous endpoint root.

Proof. Write $\varepsilon = \lambda + 2\alpha/n$. Equation (6.25) turns the pole-cleared secular equation into

$$\varepsilon\left(1 + \frac{2b}{n}K_n(\lambda)\right) + \frac{2b}{n^2} = 0. \tag{6.44}$$

Let C_0 be the bound for K_n. On $|\varepsilon + 2b/n^2| = Rn^{-3}$, the difference between the left side and $\varepsilon + 2b/n^2$ is at most

$$\frac{2|b|C_0}{n}\left(\frac{2|b|}{n^2} + \frac{R}{n^3}\right) \leq C_1 n^{-3}$$

for $n \geq R$, where C_1 is independent of R. Choose $R > C_1$ and apply Rouché's theorem to obtain the unique zero and (6.43).

On the endpoint disk write $H^{(0)}_{n,-}(\lambda) = \varepsilon J_n(\lambda)$, with J_n holomorphic and nonvanishing. Since $G_n = 1/(n\varepsilon) + K_n$ and $\delta = b/n$,

$$Q_{n,-}(\lambda) = J_n(\lambda)\left[\varepsilon\left(1 + \frac{2b}{n}K_n(\lambda)\right) + \frac{2b}{n^2}\right].$$

For $b \neq 0$ the bracket is nonzero at $\varepsilon = 0$, so the homogeneous pole is canceled. For $b = 0$ the bracket is ε, so the endpoint root is retained. In either case the counted zero is the unique zero of $Q_{n,-}$ in the disk. ∎

Definition 6.15 (Critical first cluster)**.** Fix $\alpha > 0$ and $\beta \geq 0$, set $\omega = \alpha/n$, $\omega_0 = \beta/n$, and put $k = 2\pi$. For a fixed $A > 0$, let $\mathcal{C}_{1,n}(A)$ contain, with algebraic multiplicity, the protected roots $p^{\pm}_{1,n}$ and, if $\alpha \neq k$, the affected roots in the two disks $|\lambda - p^{\pm}_{1,n}| < An^{-2}$. If $\alpha = k$, instead include all affected roots in the disk centered at $\lambda_c = \cos(k/n) - 1 - k/n$ given by

$$|\lambda - \lambda_c| < An^{-3/2}.$$

Lemma 6.16 proves that there is $A_0 = A_0(\alpha, \beta)$ such that, for each fixed $A \geq A_0$, $\mathcal{C}_{1,n}(A) = \mathcal{C}_{1,n}(A_0)$ for all sufficiently large n (with the threshold allowed to depend on A). This stabilized multiset is the critical first cluster.

Lemma 6.16 *(Critical right-edge exhaustion)***.** *Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, and put $k = 2\pi$. There is $A_0 = A_0(\alpha, \beta) > 0$ such that, for every fixed $A \geq A_0$, $\mathcal{C}_{1,n}(A) = \mathcal{C}_{1,n}(A_0)$ for all sufficiently large n. Every nonzero eigenvalue with maximal real part belongs to this critical first cluster. More precisely, if $\Lambda_{1,n}$ is the largest real part within that cluster, then every nonzero eigenvalue outside the cluster satisfies*

$$\Re\lambda \leq \Lambda_{1,n} - \begin{cases} cn^{-2}, & \alpha \leq k, \\ cn^{-1}, & \alpha > k, \end{cases} \tag{6.45}$$

for a constant $c = c(\alpha, \beta) > 0$.

Proof. The protected roots are the homogeneous roots from Definition 6.9. Put $b = \beta - \alpha$. The argument below includes $b = 0$; in that case the affected roots coincide with their homogeneous odd-sector counterparts, and Lemmas 6.13 and 6.14 give the same local counts with zero shift.

Gershgorin's theorem confines all eigenvalues to a fixed disk; choose R so that this disk lies in $|\lambda| \leq R$ and use Lemma 6.12 with that R. The constants are chosen in the following order.

1. If $\alpha \leq k$, choose a midpoint value B first so that the protected first root lies inside $\Omega_{B,n}$ with an n^{-2}-scale margin and then so that $2|b|C/\sqrt{B} < 1/8$. If $\alpha > k$, fix once and for all a $d \in (d_1, d_2)$ and use $\Omega_{d,n}$.

2. Choose A_0 so that $2|b|C_B/A_0 < 1/8$ in the first case or $2|b|C_d/A_0 < 1/4$ in the second. Increase A_0, if necessary, so that it exceeds the finitely many simple-pole thresholds and, when $\alpha = k$, the grouped threshold in Lemma 6.13. When $\alpha > k$, include the threshold for the fast first pole as well.

3. Fix any $A \geq A_0$. Only now choose n large enough for the estimates with this A and, additionally, for $2|b|C_B/n < 1/4$ or $2|b|C_d/n < 1/4$, respectively.

Since $2\delta = 2b/n$, item 1 of the off-pole estimate gives explicitly

$$|2\delta G_n| \leq \frac{2|b|C_B}{n} + \frac{2|b|C_B}{A} + \frac{2|b|C}{\sqrt{B}} < \frac{1}{2}$$

when $\alpha \leq k$. Item 2 gives

$$|2\delta G_n| \leq \frac{2|b|C_d}{n} + \frac{2|b|C_d}{A} < \frac{1}{2}$$

when $\alpha > k$. Thus in either case

$$|2\delta G_n(\lambda)| < \frac{1}{2}. \tag{6.46}$$

Thus every affected zero in the chosen right-edge window lies in one of the pole neighborhoods specified in that lemma.

Apply Lemma 6.13 to the finitely many simple pole neighborhoods in the chosen window. Each contains exactly one affected root, with expansion (6.40). When $\alpha = k$, apply its grouped assertion to the first neighborhood; together with (6.46), these counts account for every affected root in the window. When $\alpha > k$, the window contains only the slow first pole, whereas Definition 6.15 also contains the fast first disk. The simple-pole assertion of the same lemma supplies its unique affected root.

For $\alpha \neq k$, applying the simple-pole count once with A_0 and once with A shows that the annular difference between the two first disks contains no affected root. For $\alpha = k$, the corresponding statement is part of the grouped count. The protected roots do not depend on A, and hence $\mathcal{C}_{1,n}(A) = \mathcal{C}_{1,n}(A_0)$ with algebraic multiplicity. This proves the stabilization asserted in Definition 6.15.

Whenever $\ell \geq 1$ is fixed and $\alpha < \ell k$, the two poles are nonreal conjugates and

$$\Re R^{\pm}_{\ell,n} = \frac{1}{2}, \qquad \Re p^{\pm}_{\ell,n} = -\frac{\alpha}{n} - \frac{\ell^2 k^2}{2n^2} + O(n^{-4}). \tag{6.47}$$

Suppose first that $\alpha < k$. Then equations (6.40)– (6.47) show that the protected-versus-affected real-part comparison is the same for every fixed ℓ, and that the $\ell = 2$ cluster is to the left of the first by $3k^2/(2n^2) + O(n^{-3})$. We now make the global partition explicit. The midpoint definition of B in Lemma 6.11 puts precisely the fixed-index nonzero-wave poles with $1 \leq \ell \leq L$ in $\Omega_{B,n}$ for all large n. Lemma 6.13 gives one affected root at each of their simple poles, while (6.46) leaves no affected root on the complement of those disks. Among the finitely many clusters with $2 \leq \ell \leq L$, the preceding comparison puts every selected root at least $c_1 n^{-2}$ to the left of the selected first cluster. Every remaining nonzero-wave root is outside $\Omega_{B,n}$; because the protected first root lies inside that window with the margin fixed in item 1 of the constant choice, such a root is at least $c_2 n^{-2}$ to the left of the largest real part in the first cluster. Thus the nonzero-wave roots outside the first cluster satisfy (6.45) with $c = \min(c_1, c_2)$.

For every fixed $\alpha > 0$, the zero-wave endpoint is the only unpaired affected pole on the $O(n^{-1})$ scale; the $q = \pi$ endpoint, when present, stays a fixed distance into the left half-plane. Lemma 6.14 gives its unique affected root and places its real part at $-2\alpha/n + O(n^{-2})$. For $\alpha < k$ this is a distance of order n^{-1} to the left of the first cluster. Combined with the nonzero-wave partition above, this proves (6.45) for $\alpha < k$.

At $\alpha = k$, the first protected cluster contains

$$-\frac{k}{n} + \left(-\frac{k^2}{2} + \frac{k^2}{\sqrt{3}}\right) n^{-2} + O(n^{-3}). \tag{6.48}$$

For every fixed $\ell \geq 2$, equations (6.40)– (6.47) apply with $\alpha = k$. If $\beta \leq k$, its largest possible real part is below (6.48) by

$$\left[\frac{(\ell^2-1)k^2}{2} + \frac{k^2}{\sqrt{3}} - 2(k-\beta)\right] n^{-2} + O(n^{-3}),$$

which is positive because $k > 2\sqrt{3}$. If $\beta > k$, the affected higher modes move to the left and the protected comparison is even stronger. Estimate (6.46) accounts for all remaining affected roots. The endpoint formula (6.43) from Lemma 6.14 puts the zero-wave affected root at $-2k/n + O(n^{-2})$ and hence a distance of order n^{-1} to the left of the first cluster. This proves the assertion at $\alpha = k$.

Finally, if $\alpha > k$, the first slow pole has decay coefficient $d_1 = \alpha - \sqrt{\alpha^2 - k^2}$. By the choice of d, the only protected pole in $\Omega_{d,n}$ is the first slow pole, and (6.46)– (6.40) show that its affected partner is the only affected root there. Both have real part $-d_1/n + O(n^{-2})$, whereas every nonzero root outside the window has real part at most $-d/n$. In particular, the endpoint root (6.43) is at $-2\alpha/n + O(n^{-2})$ and lies outside the window because $d < 2\alpha$. This gives a separation of order n^{-1} and completes the proof. ■

Theorem 6.17 *(Bulk-critical double scaling)***.** *Set*

$$\omega = \frac{\alpha}{n}, \qquad \omega_0 = \frac{\beta}{n}, \qquad k = 2\pi,$$

where $\alpha > 0$ and $\beta \geq 0$ are fixed. Write $\gamma_n = \gamma_n(\alpha/n, \beta/n)$ for the gap in Definition 6.2. Away from $\alpha = k$,

$$\alpha < k: \quad \gamma_n = \frac{\alpha}{n} + \left[\frac{k^2}{2} - 2(\alpha-\beta)_+\right] n^{-2} + O(n^{-3}), \tag{6.49}$$

$$\begin{aligned} \alpha > k: \quad \gamma_n = {} & \frac{\alpha - \sqrt{\alpha^2-k^2}}{n} \\ & + \left[\frac{k^2}{2} - 2(\beta-\alpha)_+ \frac{\alpha - \sqrt{\alpha^2-k^2}}{\sqrt{\alpha^2-k^2}}\right] n^{-2} + O(n^{-3}). \end{aligned} \tag{6.50}$$

At $\alpha = k$,

$$\beta \le k : \quad \gamma_n = \frac{k}{n} + k^2\left(\frac{1}{2} - \frac{1}{\sqrt{3}}\right) n^{-2} + O(n^{-3}), \tag{6.51}$$

$$\beta > k : \quad \gamma_n = \frac{k}{n} - 2\sqrt{k(\beta - k)}\, n^{-3/2} + O(n^{-2}). \tag{6.52}$$

The constants in the $O(\cdot)$ terms may depend on the fixed pair (α, β). In particular, (6.49) and (6.50) are side expansions for fixed $\alpha \ne k$ and are not uniform as $\alpha \to k$; the exact scaling $\alpha = k$ is the separate regime (6.51)– (6.52). No uniformity as $\alpha \downarrow 0$ is asserted for either the remainder constants or the lower bound on n. Thus the first homogeneous exceptional scaling is a square-root boundary layer for a faster defect, while the side selecting the affected parity mode reverses as α passes through 2π.

Proof. For $\alpha < k$, the protected first pair has

$$\Re\lambda_1^+ = -\frac{\alpha}{n} - \frac{k^2}{2n^2} + O(n^{-3}).$$

The pole residue tends to $R_1 = \frac{1}{2} + \mathrm{i}\alpha/(2\sqrt{k^2 - \alpha^2})$, and $\delta = (\beta - \alpha)/n$. Hence the affected real part shifts by $-2(\beta - \alpha)n^{-2} + O(n^{-3})$ by (6.40). Taking the larger real part gives (6.49).

For $\alpha > k$, put $s = \sqrt{\alpha^2 - k^2}$. The protected rate is $(\alpha - s)n^{-1} + k^2(2n^2)^{-1} + O(n^{-3})$, while $R_1 \to -(\alpha - s)/(2s)$. The same pole equation shifts the affected eigenvalue by $2(\beta - \alpha)(\alpha - s)/(sn^2) + O(n^{-3})$. Selection of the larger real part gives (6.50). Lemma 6.16 shows in both cases that no root outside the first cluster can compete with the selected root.

At $\alpha = k$, the exact lattice splitting of the homogeneous first block is

$$\sqrt{\omega^2 - \sin^2(k/n)} = \frac{k^2}{\sqrt{3}\, n^2} + O(n^{-4}),$$

which proves (6.51) when the protected root wins. Near the center of this block, put

$$\lambda_c = \cos(k/n) - 1 - k/n, \qquad r_n = \sqrt{(k/n)^2 - \sin^2(k/n)}, \qquad \Delta = \lambda - \lambda_c.$$

Lemma 6.10 gives the uniform identity

$$G_n(\lambda) = \frac{2(\Delta - k/n)}{n(\Delta^2 - r_n^2)} + H_n(\lambda), \qquad |H_n(\lambda)| \le C, \tag{6.53}$$

throughout a fixed $n\Delta$-neighborhood of zero. In particular this bound is uniform on every contour used below.

For $\beta > k$, put $b = \beta - k$, $\epsilon = n^{-1/2}$, and $\Delta = n^{-3/2}Y$. Since

$$n^4 r_n^2 = \frac{k^4}{3} + O(n^{-2}),$$

the secular equation, uniformly for Y in a compact set bounded away from zero, is

$$F_n(Y) = 1 - \frac{4kb}{Y^2} + \frac{4b\epsilon}{Y} + O(\epsilon^2) = 0. \tag{6.54}$$

The limiting function has simple zeros $Y_\pm = \pm 2\sqrt{kb}$. On $|Y - Y_\pm| = C\epsilon$, its modulus is bounded below by $cC\epsilon$, while the difference in (6.54) is at most $C_1\epsilon$. Choosing $C > C_1/c$, Rouché's theorem gives one zero in each disk and hence two affected roots

$$\Delta = \pm 2\sqrt{kb}\, n^{-3/2} + O(n^{-2}).$$

The positive displacement gives (6.52).

For $\beta < k$, put $h = k - \beta > 0$. With the same scaling, the secular equation is

$$\widetilde{F}_n(Y) = 1 + \frac{4kh}{Y^2} - \frac{4h\epsilon}{Y} + O(\epsilon^2) = 0. \tag{6.55}$$

Its two leading zeros are $Y_\pm = \pm 2\mathrm{i}\sqrt{kh}$. Substitution shows that $Y_\pm + 2h\epsilon$ has residual $O(\epsilon^2)$. On the circles $|Y - (Y_\pm + 2h\epsilon)| = C\epsilon^2$, the affine part has modulus at least $cC\epsilon^2$ and the remaining error is at most $C_1\epsilon^2$. A second application of Rouché's theorem therefore gives the affected pair

$$\Delta = \pm 2\mathrm{i}\sqrt{kh}\, n^{-3/2} + 2hn^{-2} + O(n^{-5/2}). \tag{6.56}$$

Its decay rate exceeds the protected decay rate by

$$\left(\frac{k^2}{\sqrt{3}} - 2h\right) n^{-2} + O(n^{-5/2}).$$

This coefficient is positive for $0 < h \le k$ because $k = 2\pi > 2\sqrt{3}$. The protected root therefore wins for every $\beta < k$; at $\beta = k$ the two parity roots coincide. This proves (6.51); the case $\beta > k$ was obtained from the positive displacement above. Lemma 6.16 supplies the required exclusion of every higher, endpoint, and off-pole affected root in both cases. ∎

Table 2 collects the gap-controlling branch and the first correction scale from Theorem 6.17. An equality case in the second column means that the protected and affected parity roots coincide.

Regime	Gap-controlling parity	Leading gap	First correction
$\alpha < k$	affected if $\beta < \alpha$; protected if $\beta > \alpha$; tie if $\beta = \alpha$	αn^{-1}	n^{-2}, (6.49)
$\alpha = k$, $\beta \le k$	protected if $\beta < k$; tie if $\beta = k$	kn^{-1}	n^{-2}, (6.51)
$\alpha = k$, $\beta > k$	affected	kn^{-1}	$n^{-3/2}$, (6.52)
$\alpha > k$	protected if $\beta < \alpha$; affected if $\beta > \alpha$; tie if $\beta = \alpha$	$(\alpha - \sqrt{\alpha^2 - k^2})n^{-1}$	n^{-2}, (6.50)

Table 2: Selection and correction scales in the bulk-critical limit, with $k = 2\pi$.

Figure 1 displays the branch selection in the (α, β) plane. The diagonal is the equal-rate locus, whereas the vertical line is the first homogeneous exceptional scaling. Their different roles account for the reversal of the affected side across $\alpha = 2\pi$.

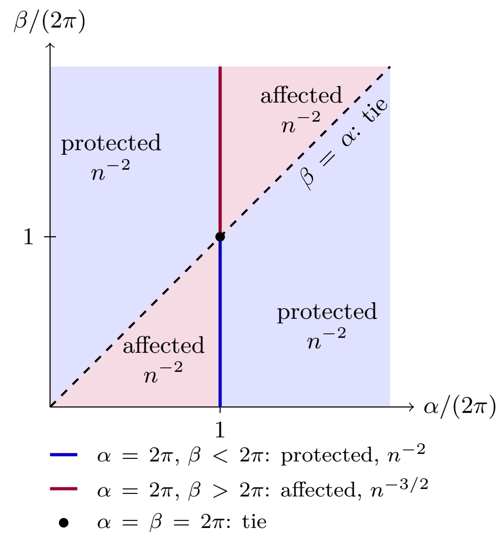

Figure 1: Gap-controlling parity and first correction scale in the bulk-critical limit. Region labels apply away from $\alpha = 2\pi$; the colored segments on that line give its separate boundary regime.

7. Telegraph comparison

The standard one-dimensional telegraph system follows from the right- and left-moving continuum equations in [Sch93, Sec. II, Eqs. (2.2)–(2.4)]. In that convention a particle reverses direction at one half of the collision rate. Setting the speed to one and identifying that collision rate with 2ω gives the normalization below. We compare it with the next lattice correction before introducing an independently defined continuum point interaction at the same unit site spacing. No convergence of lattice operators under a vanishing-spacing limit is claimed.

Definition 7.1 (Two continuum approximations)**.** For a homogeneous ring of circumference n, the standard telegraph system is

$$\partial_t \rho = -\partial_x m, \qquad \partial_t m = -\partial_x \rho - 2\omega m. \tag{7.1}$$

Its slow Fourier eigenvalue at wave number q is

$$\lambda^+_{\mathrm{tel}}(q) = -\omega + \sqrt{\omega^2 - q^2}. \tag{7.2}$$

Here and below the square root has nonnegative real part and, when it is purely imaginary, nonnegative imaginary part. The lattice-diffusion-corrected system is

$$\partial_t \rho = -\partial_x m + \tfrac{1}{2}\partial_x^2 \rho, \qquad \partial_t m = -\partial_x \rho + \tfrac{1}{2}\partial_x^2 m - 2\omega m. \tag{7.3}$$

Its slow eigenvalue is

$$\lambda^+_{\mathrm{td}}(q) = -\frac{q^2}{2} - \omega + \sqrt{\omega^2 - q^2}. \tag{7.4}$$

Lemma 7.2 *(Second-order lattice expansion)***.** *Let $\rho, m \in C^4(\mathbb{R}/n\mathbb{Z})$ and sample $u_j^\pm = \{\rho(j) \pm m(j)\}/2$. The homogeneous lattice generator induces*

$$\begin{aligned} \dot\rho_j &= -m'(j) + \tfrac{1}{2}\rho''(j) + r_{\rho,j}, \\ \dot m_j &= -\rho'(j) + \tfrac{1}{2}m''(j) - 2\omega m(j) + r_{m,j}, \end{aligned} \tag{7.5}$$

where

$$\begin{aligned} |r_{\rho,j}| &\le \tfrac{1}{6}\|m'''\|_\infty + \tfrac{1}{24}\|\rho^{(4)}\|_\infty, \\ |r_{m,j}| &\le \tfrac{1}{6}\|\rho'''\|_\infty + \tfrac{1}{24}\|m^{(4)}\|_\infty. \end{aligned} \tag{7.6}$$

Thus (7.3) is the second-order long-wave truncation of the lattice equations, with the remainder controlled in the sampled ℓ^∞ norm by (7.6).

Proof. Adding and subtracting the two homogeneous equations (2.1)–(2.2) gives the exact sampled identities

$$\begin{aligned} \dot\rho_j &= \frac{\rho(j-1)+\rho(j+1)}{2} - \rho(j) + \frac{m(j-1)-m(j+1)}{2}, \\ \dot m_j &= \frac{\rho(j-1)-\rho(j+1)}{2} + \frac{m(j-1)+m(j+1)}{2} - m(j) - 2\omega m(j). \end{aligned}$$

Taylor's theorem on the unit intervals adjacent to j bounds the error in each centered first difference by one sixth of the supremum of the third derivative, and the error in each centered second difference by one twenty-fourth of the supremum of the fourth derivative. Substitution gives (7.5)–(7.6). ∎

Theorem 7.3 *(Exact relaxation rate versus telegraph rates)***.** *Fix $\omega > 0$, let $q = 2\pi/n$, and define $\gamma_{\mathrm{tel}} = -\Re\lambda^+_{\mathrm{tel}}(q)$ and $\gamma_{\mathrm{td}} = -\Re\lambda^+_{\mathrm{td}}(q)$. Then, as $n \to \infty$,*

$$\gamma_n^{\mathrm{hom}} - \gamma_{\mathrm{tel}} = \frac{q^2}{2} - \left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6), \tag{7.7}$$

$$\gamma_n^{\mathrm{hom}} - \gamma_{\mathrm{td}} = -\left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6). \tag{7.8}$$

Consequently

$$\frac{\gamma_{\mathrm{tel}}}{\gamma_n^{\mathrm{hom}}} \longrightarrow \frac{1}{1+\omega}, \qquad \frac{\gamma_{\mathrm{td}}}{\gamma_n^{\mathrm{hom}}} \longrightarrow 1. \tag{7.9}$$

Separately, let $\alpha > 0$ be fixed and set $\omega = \alpha/n$. The standard telegraph decay rate is exactly

$$\gamma_{\mathrm{tel}} = \frac{\alpha - \Re\sqrt{\alpha^2 - (2\pi)^2}}{n}, \tag{7.10}$$

which agrees with the coefficient of n^{-1} in (6.49)–(6.52). For fixed $\beta \ge 0$, the defect-dependent corrections under $\omega_0 = \beta/n$, of orders n^{-2} and $n^{-3/2}$ in those formulas, are beyond this homogeneous telegraph comparison.

Proof. The exact lattice dispersion replaces $-q^2/2$ by $\cos q - 1$ and q inside the square root by $\sin q$. Expanding $\cos q$, $\sin^2 q$, and the square root gives (7.7)–(7.8). Dividing by the leading exact rate $(1+\omega)q^2/(2\omega)$ gives (7.9). Substitution of $\omega = \alpha/n$ and $q = 2\pi/n$ into (7.2), followed by taking the negative real part, gives (7.10). ∎

Proposition 7.4 *(Point-defect telegraph analogue and secular equation)***.** *Let $\delta \in \mathbb{R}$. For the kinetic defect in Definition 2.1, one has $\delta = \omega_0 - \omega \geq -\omega$. Here the continuum coordinate uses the same unit site spacing as the lattice, so the circle has circumference n. The Dirac coupling has coefficient 2δ and enters the polarization equation as $-2\delta\, \delta_0(x)m$; no further lattice-spacing limit or rescaling of δ is imposed. On the circle cut at zero, define the point-defect telegraph operator by*

$$\mathcal{A}_\delta(\rho, m) = (-m', -\rho' - 2\omega m) \quad \text{on } (0, n),$$

with domain consisting of $m, \rho \in H^1(0,n)$ having traces that satisfy

$$m(0+) = m(n-), \qquad \rho(0+) - \rho(n-) = -2\delta m(0). \tag{7.11}$$

This is the distributional realization obtained by adding $-2\delta\, \delta_0(x)m$ to the second equation in (7.1). Define the homogeneous telegraph spectrum on this circle by

$$\Sigma_{\mathrm{tel}}(n,\omega) = \left\{-\omega \pm \sqrt{\omega^2 - (2\pi\ell/n)^2} \,:\, \ell \in \mathbb{Z}\right\}.$$

For $\lambda \notin \Sigma_{\mathrm{tel}}(n,\omega)$, put

$$\kappa = \sqrt{\lambda(\lambda + 2\omega)}$$

and interpret $\kappa^{-1}\coth(n\kappa/2)$ as the even meromorphic function of κ obtained by analytic continuation; no sign choice for the square root is required. The affected-parity eigenvalues outside $\Sigma_{\mathrm{tel}}(n,\omega)$ are exactly the zeros of

$$1 + \delta\frac{\lambda}{\kappa}\coth\left(\frac{n\kappa}{2}\right) = 0. \tag{7.12}$$

For the involution

$$\mathcal{R}(\rho, m)(x) = (\rho(n-x), -m(n-x)),$$

the reflection-even telegraph modes are unchanged. Formula (7.12) is the continuum analogue of (3.8) at unit site spacing; it is not an operator-limit statement. The exact lattice formula retains $1 - \cos q$ and $\sin q$, and (7.7)–(7.8) quantify the resulting relaxation-rate error.

Proof. The jump in (7.11) cancels the delta distribution in $-\rho' - 2\delta\delta_0 m(0)$, while continuity of m prevents a delta distribution in $-m'$. Thus the stated domain gives the point interaction on $L^2((0,n);\mathbb{C}^2)$. It is densely defined because it contains $C_c^\infty((0,n);\mathbb{C}^2)$, and it is closed: the graph norm controls $\|m'\|_2$ directly and controls $\|\rho'\|_2$ by $\|\rho'\|_2 \leq \|(\mathcal{A}_\delta(\rho,m))_2\|_2 + 2\omega\|m\|_2$; the trace theorem then preserves (7.11) under graph-norm limits.

For $\delta = 0$, substitution of the periodic Fourier mode $e^{2\pi i \ell x/n}$ gives $\lambda(\lambda + 2\omega) + (2\pi\ell/n)^2 = 0$. Hence its two roots, as ℓ ranges over $\mathbb{Z}$, are exactly $\Sigma_{\mathrm{tel}}(n,\omega)$. Indeed, the periodic Fourier transform identifies the homogeneous operator, with its periodic H^1 domain, with the orthogonal direct sum over $\ell \in \mathbb{Z}$ of the matrices

$$M_\ell := \begin{pmatrix} 0 & -\mathrm{i}q_\ell \\ -\mathrm{i}q_\ell & -2\omega \end{pmatrix}, \qquad q_\ell = \frac{2\pi\ell}{n}.$$

Their characteristic polynomials are $\lambda(\lambda+2\omega)+q_\ell^2$. To see that the direct sum adds no further spectrum, fix ζ outside this root set. Its inverse block is

$$(\zeta I - M_\ell)^{-1} = \frac{1}{\zeta(\zeta+2\omega)+q_\ell^2}\begin{pmatrix} \zeta+2\omega & -\mathrm{i}q_\ell \\ -\mathrm{i}q_\ell & \zeta \end{pmatrix}.$$

For all sufficiently large $|\ell|$, the denominator has modulus at least $q_\ell^2/2$, so $\|(\zeta I - M_\ell)^{-1}\| \le C_\zeta/(1+|q_\ell|)$; the finitely many remaining inverse blocks are bounded as well. Hence their direct sum maps L^2 boundedly into the weighted Fourier domain corresponding to periodic H^1, and is the resolvent of the homogeneous operator. Thus the displayed roots exhaust its spectrum.

Let $\lambda \notin \Sigma_{\mathrm{tel}}(n,\omega)$. The eigenvalue equations away from zero are

$$-m' = \lambda\rho, \qquad -\rho' - 2\omega m = \lambda m.$$

Thus $\lambda \neq 0$ and elimination of ρ gives $m'' = \kappa^2 m$. Write

$$m(x) = A\cosh(\kappa(x-n/2)) + B\sinh(\kappa(x-n/2)).$$

The equality $m(0) = m(n)$ gives $B = 0$, since $\sinh(n\kappa/2) = 0$ would put λ in the homogeneous spectrum. Using $\rho = -m'/\lambda$, the second jump condition in (7.11) is equivalent to

$$m'(0) - m'(n) = 2\delta\lambda m(0). \tag{7.13}$$

Substitution of the even hyperbolic-cosine solution into (7.13) gives

$$-2\kappa\sinh(n\kappa/2) = 2\delta\lambda\cosh(n\kappa/2),$$

which is exactly (7.12). Conversely, if (7.12) holds outside the homogeneous spectrum, take $A = 1$, $B = 0$ in the displayed hyperbolic solution and put $\rho = -m'/\lambda$. Equation (7.13) and the continuity of m then give both conditions in (7.11), so this construction is a nonzero eigenfunction. Moreover $m(n-x) = m(x)$ and $\rho(n-x) = -\rho(x)$, hence $\mathcal{R}(\rho,m) = -(\rho,m)$ and the eigenfunction has affected parity.

Equivalently, the polarization entry of the homogeneous telegraph resolvent is

$$\frac{1}{n}\sum_{\ell\in\mathbb{Z}} \frac{\lambda}{\lambda(\lambda+2\omega)+(2\pi\ell/n)^2} = \frac{\lambda}{2\kappa}\coth\left(\frac{n\kappa}{2}\right).$$

The series converges normally away from the homogeneous spectrum and the last expression supplies its meromorphic continuation when κ is purely imaginary, in agreement with the direct jump calculation. Finally, a reflection-even vector satisfies $m(0+) = -m(n-)$; continuity of m therefore gives $m(0) = 0$. The jump vanishes, so these modes are protected exactly as in Lemma 4.2. ∎

8. Conclusions

Remark 8.1 (The theorem package). For the generator in Definition 2.1:

1. Theorem 3.3 and Theorem 3.4 give exact characteristic equations for every n, ω, and ω_0; the trace equation is used away from its denominators and its cleared polynomial identity is used at their zeros.

2. Theorems 4.8 and 4.9 classify the algebraic and geometric multiplicities of every eigenvalue, including every exceptional point.

3. Theorem 5.3 gives an exact criterion and decay length for a defect-localized mode, and Proposition 5.4 gives its finite-ring convergence.

4. Theorems 6.7 and 6.17 give the fixed-rate, vanishing-defect, and bulk-critical large-n gap asymptotics.

5. Lemma 7.2, Theorem 7.3, and Proposition 7.4 derive the corrected continuum system, compare the exact relaxation rate with the standard and diffusion-corrected telegraph approximations, and construct the unit-spacing point-defect analogue without asserting an operator scaling limit.

Each item retains the hypotheses stated in its cited theorem: in particular, localization is asserted off the bulk spectrum (with the compact case treated separately), finite-ring localization convergence assumes a simple noncompact root, and the gap results use their stated fixed-rate or double-scaling regimes.

Code availability

The manuscript source and the formula-audit code are available at `https://github.com/skypher/kinetic-defect-ring`. The archived submission revision is identified by the Git tag `paper-2026-08-28-reviewed-v9`. From the repository root, the audit is run as

```
python3 -u scripts/verify_formulas.py.
```

It compares the characteristic, transfer, flat-band Jordan, localization, and gap formulas with direct finite-matrix or defining-equation calculations. In particular, it checks finite characteristic and transfer identities, flat-band nullities through the largest possible block size, selected compact and noncompact localization identities, and representative fixed-rate and critical gap expansions. It does not test the analytic off-pole root exclusions, every resultant stratum, the uniform finite-ring eigenvector bound, or the closedness and spectrum of the continuum point-interaction operator. These computations are independent checks of the displayed formulas and are not used in place of the proofs in the manuscript.

AI-use disclosure

The proof architecture and manuscript were developed with assistance from OpenAI GPT-5.6 Sol (`gpt-5.6-sol`) at max reasoning effort. The model served as an interactive research assistant for deriving and testing arguments, checking citations and calculations, writing exact verification code, and editing the exposition. The author set the research direction, evaluated and approved the final arguments, and assumes responsibility for the mathematical claims, references, computations, and text.

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